



ETRI Research Internship Final Presentation

2026-08-28

Field Robotics Laboratory | Minsoo Park

Contents

VLA pipeline

1. Pi0
2. GROOT
3. starVLA
4. RL fine-tuning

About Me



박민수 | Minsoo Park

- **Affiliation**

- SeoulTech | Department of Intelligent Robotics Engineering
 - 2021.03 ~ Present
- MoAI Lab | Undergraduate Researcher
 - 2026.01 ~ Present

- **Research Interests**

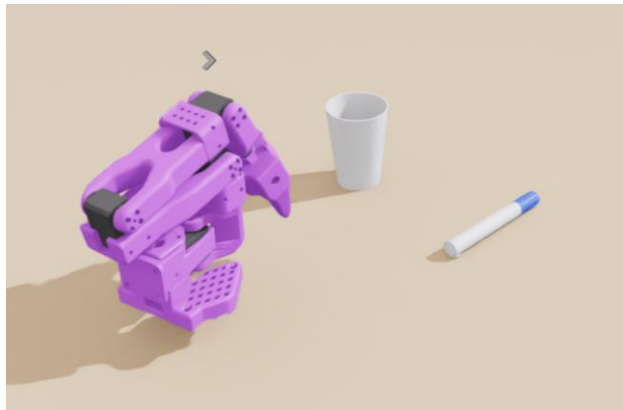
- Robot Manipulation
- Imitation Learning
- Vision-Language-Action Models

Research Goal

Can a VLA trained on simulated demonstrations be deployed on a real robot?

Task: Pick up a marker place it into a cup

Simulation



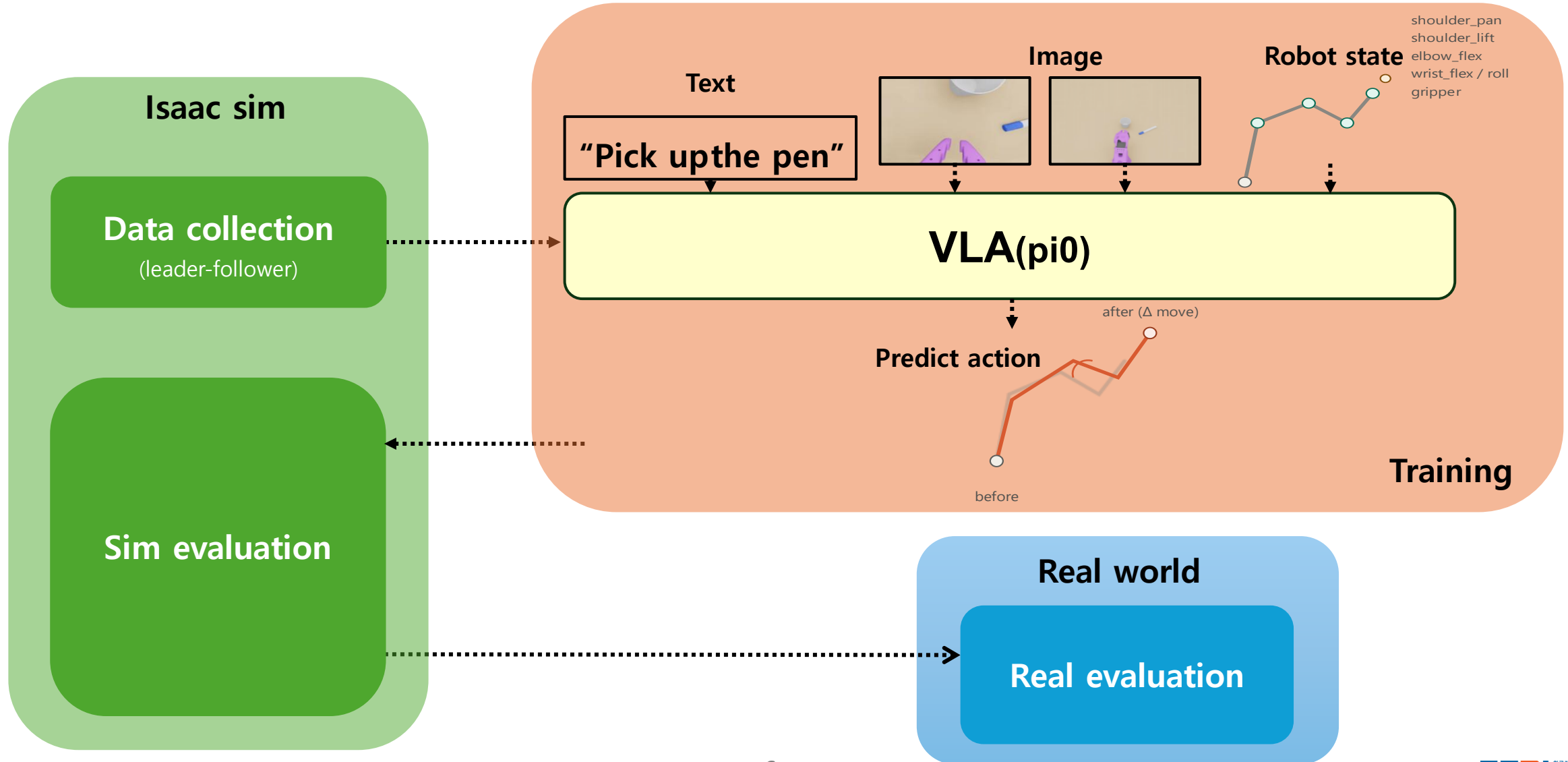
VLA Fine-tuning

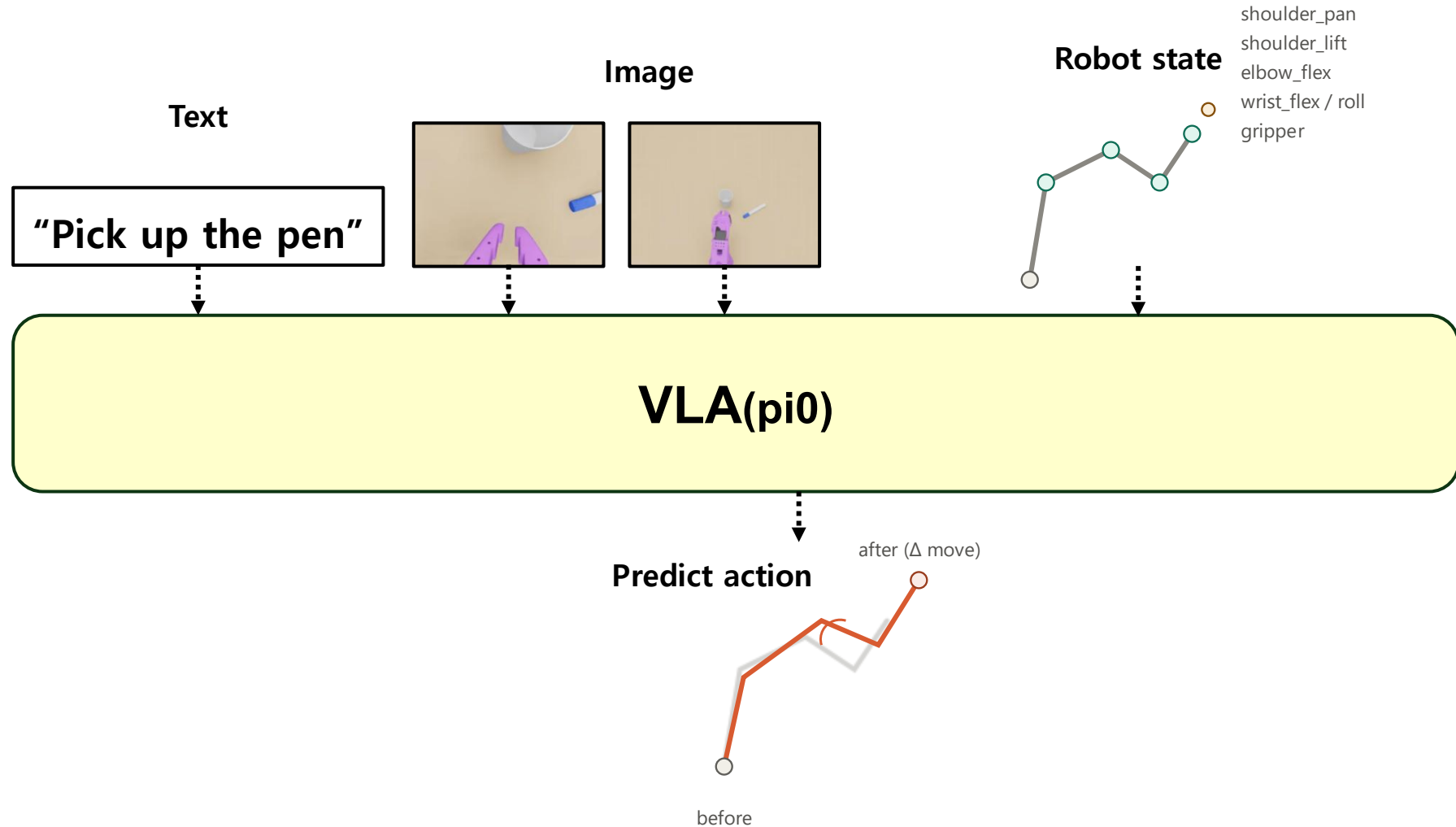
Real Robot



VLA Pipeline

Overall Research Pipeline

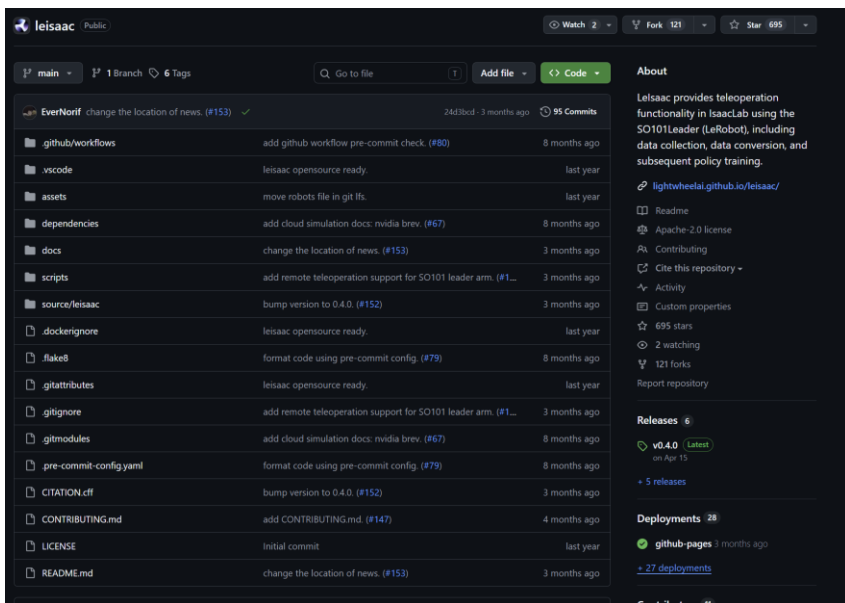




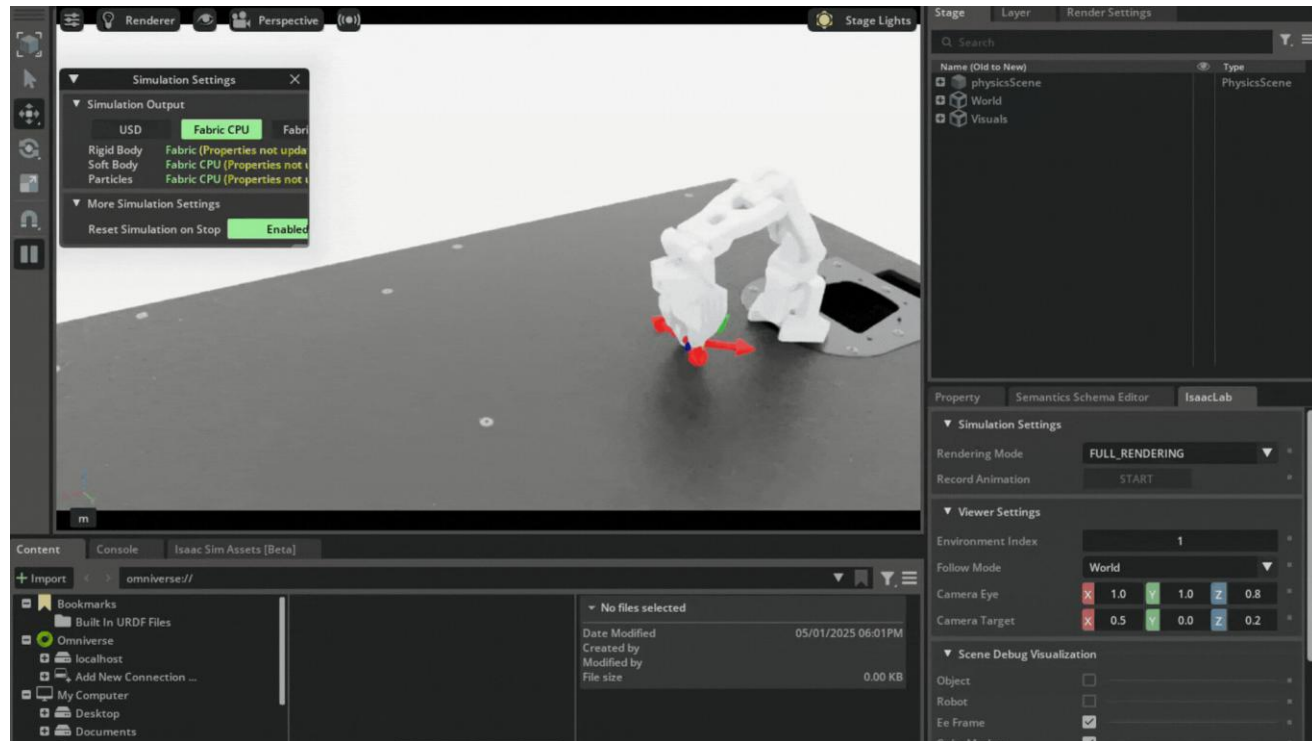
What is Isaac sim?

Isaac Sim is an advanced robotics simulation and synthetic data generation platform built on the [Nvidia Omniverse](#) framework.

Leisaac (Isaac sim + Lerobot)



- Example



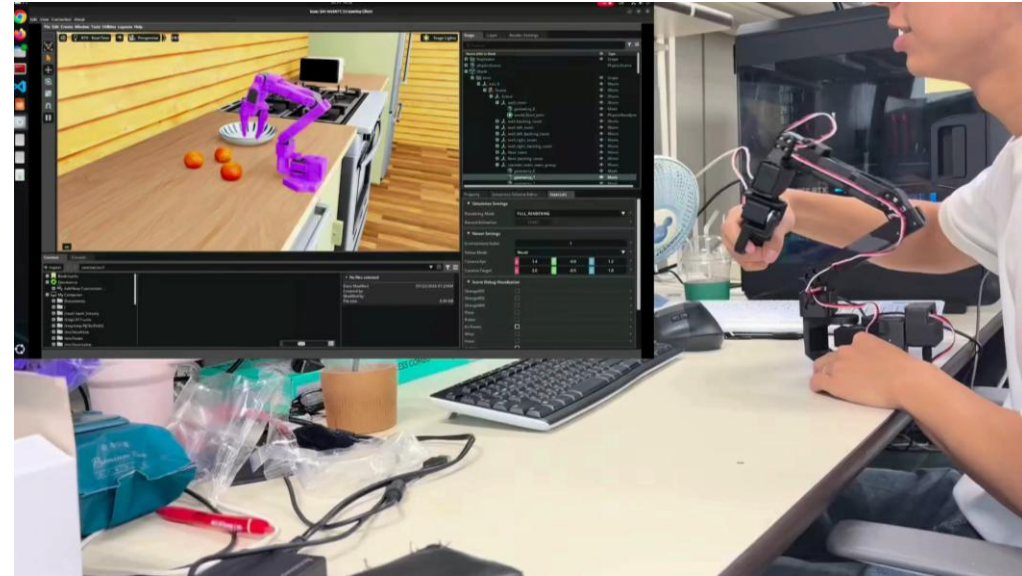
- kitchen_with_orange.zip
- Leisaac-SO101-PickOrange-v0

VLA Pipeline Data collection (isaac-sim)



Leader arm

Control



follower arm (Isaac sim)

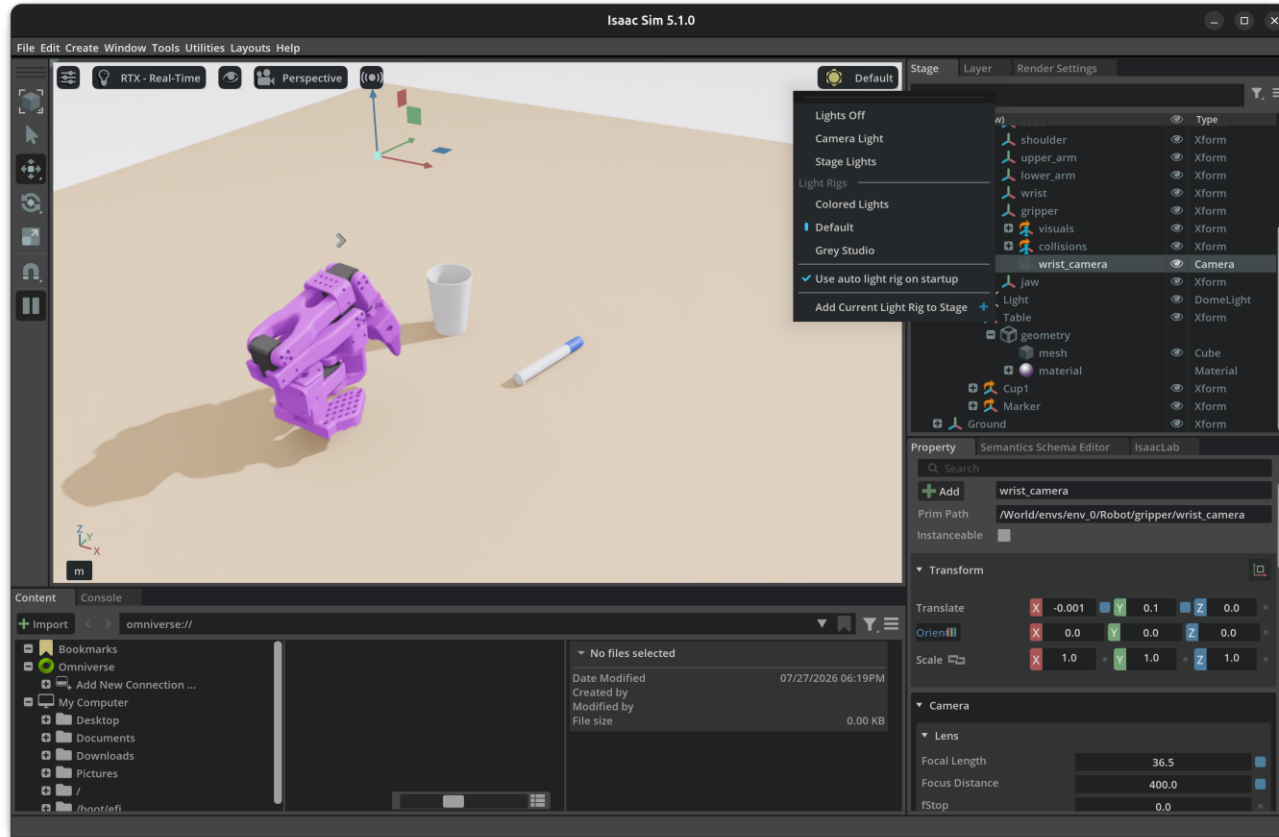
Record

Dataset

1. RGB image – 640x480, 30Hz
2. Robot state – 6D (5joint + gripper)
3. Robot action – 6D (5joint + gripper)

VLA Pipeline

Data collection (isaac-sim)



Task: pick up the marker and place into the cup

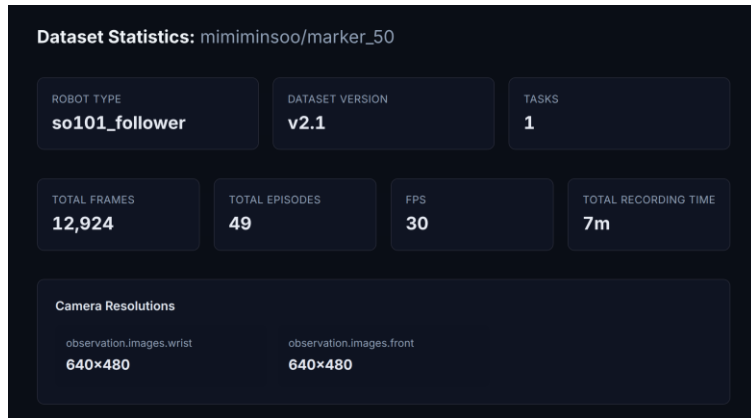
Domain randomization

-	Cup	Marker
Offset	$\pm 3\text{cm}$	$\pm 3\text{cm}$
yaw	X	360°

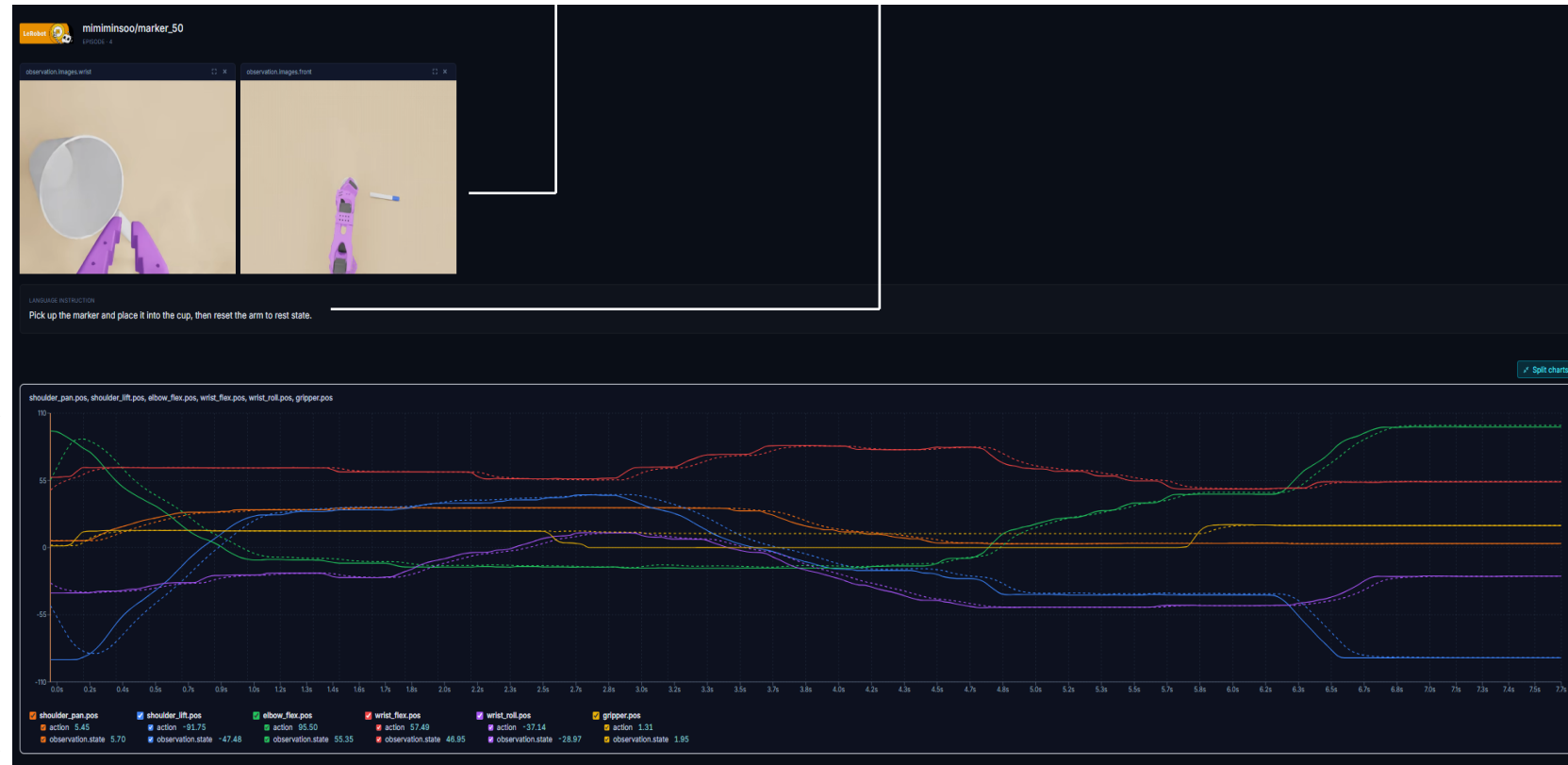
```
randomize_object_uniform(
    "cup1", pose_range={"x": (-0.03, 0.03), "y": (-0.03, 0.03), "z": (0.0, 0.0)}),
randomize_object_uniform(
    "marker",
    pose_range={"x": (-0.03, 0.03), "y": (-0.03, 0.03), "z": (0.0, 0.0),
    "yaw": (-3.14159, 3.14159)}),)
```

VLA Pipeline

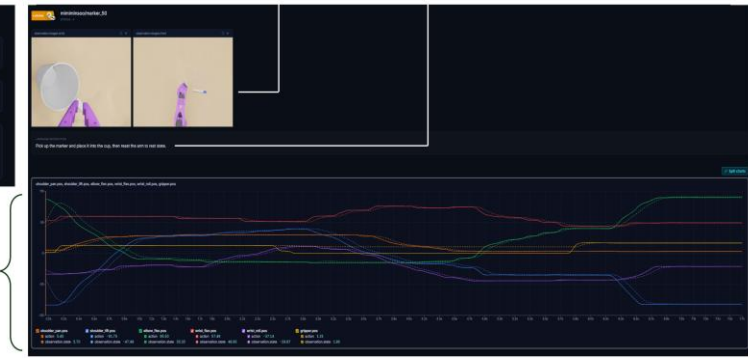
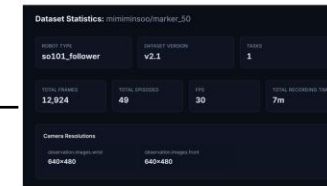
Collected dataset



Robot state & action



VLA Pipeline Training (fine-tuning)



Pi0_base

- 10000h robot data → pre-training
 - 7 robot embodiments
 - 68 task
 - Open X embodiment dataset

Fine-tuning

Pi0_LoRA_fine-tuning

- Base weights frozen
- Train only LoRA adapters
- Data: our 49 demos (7min)

Aspect	Full fine-tuning	LoRA fine-tuning
What updates	All weights	~0.1% (adapters)
Base weights	Changed	Frozen ❄️
GPU memory	70 GB+	~22.5 GB
Data needed	More (5~100H)	Less (1~5H)

Openpi config → Full fine-tuning: 100k steps (batch_size=256) ≈ 8×H100 → 2days

Measured: LoRA fine-tuning: 30k steps (batch_size =8) = RTX3090 → 24H

VLA pipeline Experiment

Complete: 
 In-progress: 
 To-do: 

1st training

Data collection
49 episode, 7min

Training
LoRA, 30K step,
Action chunk =10

Sim-evaluation
Success rate: 10%

2nd training - 1

Data collection
100 episode, 17min

Training
LoRA, 30K step,
Action chunk =10

Sim-evaluation
Success rate: **70%**

3rd training - 1

Data collection
150 episode, 24min

Training
LoRA, 30K step,
Action chunk =30

Real-evaluation
Success rate: **33%**

Sim-evaluation
Success rate: **70%**

2nd training - 2

Data collection
100 episode, 17min

Training
LoRA, 30K step,
Action chunk =30

Real-evaluation
Success rate: **25%**

Sim-evaluation
Success rate: **70%**

3rd training - 2

Data collection
180 episode, 32min

Training
LoRA, 30K step,
Action chunk =30

Real-evaluation
Success rate: **??**

Sim-evaluation
Success rate: **60%**

2nd training - 3

Data collection
100 episode, 17min

Training
LoRA, 30K step,
Action chunk =30
**Inference action
chunk = 10**

Sim-evaluation
Success rate: **33%**

VLA pipeline Experiment

50 dataset + action chunk=10

Success rate : 10%

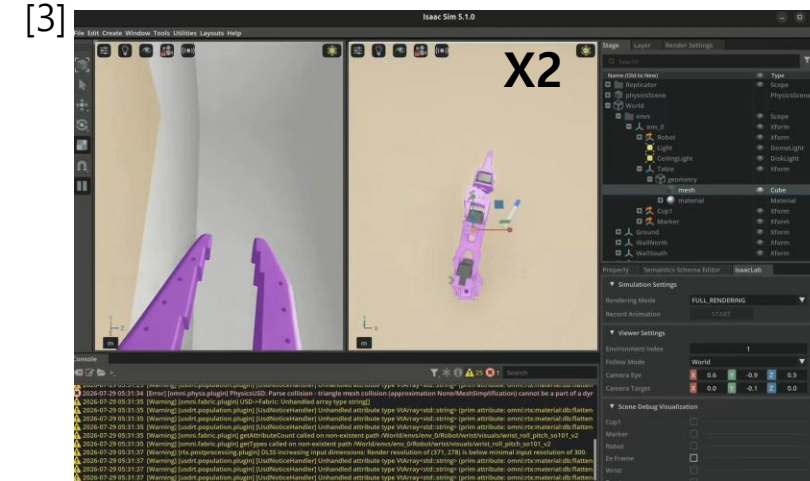
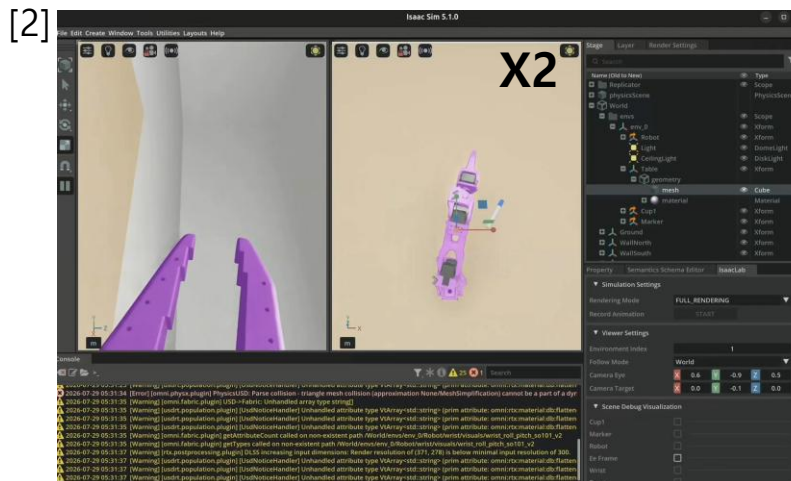
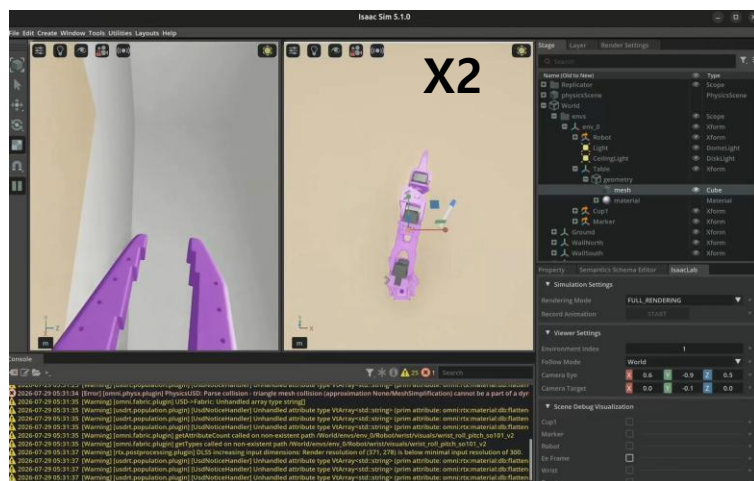
1st training

Data collection
49 episode, 7min

Training
LoRA, 30K step,
Action chunk =10

Sim-evaluation
Success rate: 10%

Complete: 
In-progress: 
To-do: 



Problem

1. Stuttering —————> Action chunk parameter
2. Can't grip the marker (almost) —————> More dataset (least 1h)

VLA pipeline Experiment

50 dataset + action chunk=10

Success rate : 0%

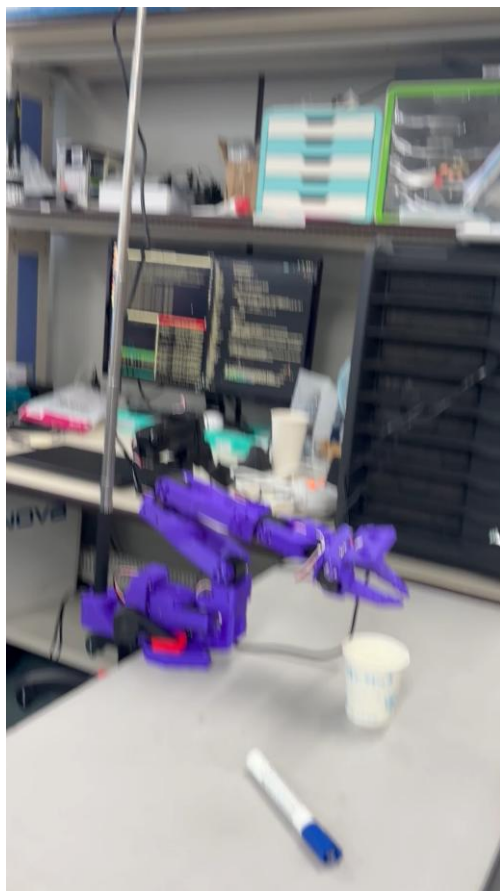
1st training

Data collection
49 episode, 7min

Training
LoRA, 30K step,
Action chunk =10

Sim-evaluation
Success rate: 10%

Complete: 
In-progress: 
To-do: 



fail



fail



fail

VLA pipeline Experiment

Complete: 
 In-progress: 
 To-do: 

1st training

Data collection
49 episode, 7min

Training
LoRA, 30K step,
Action chunk =10

Sim-evaluation
Success rate: 10%

2nd training - 1

Data collection
100 episode, 17min

Training
LoRA, 30K step,
Action chunk =10

Sim-evaluation
Success rate: **70%**

3rd training - 1

Data collection
150 episode, 24min

Training
LoRA, 30K step,
Action chunk =30

Real-evaluation
Success rate: **33%**

Sim-evaluation
Success rate: **70%**

2nd training - 2

Data collection
100 episode, 17min

Training
LoRA, 30K step,
Action chunk =30

Real-evaluation
Success rate: **25%**

Sim-evaluation
Success rate: **70%**

3rd training - 2

Data collection
180 episode, 32min

Training
LoRA, 30K step,
Action chunk =30

Real-evaluation
Success rate: **??**

Sim-evaluation
Success rate: **60%**

2nd training - 3

Data collection
100 episode, 17min

Training
LoRA, 30K step,
Action chunk =30
**Inference action
chunk = 10**

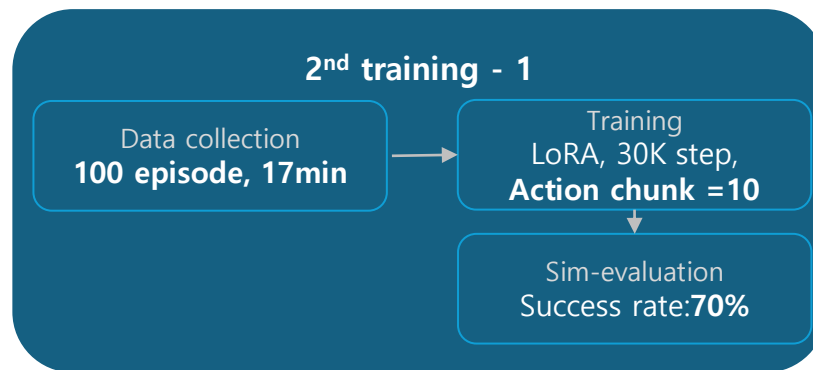
Sim-evaluation
Success rate: **33%**

VLA pipeline

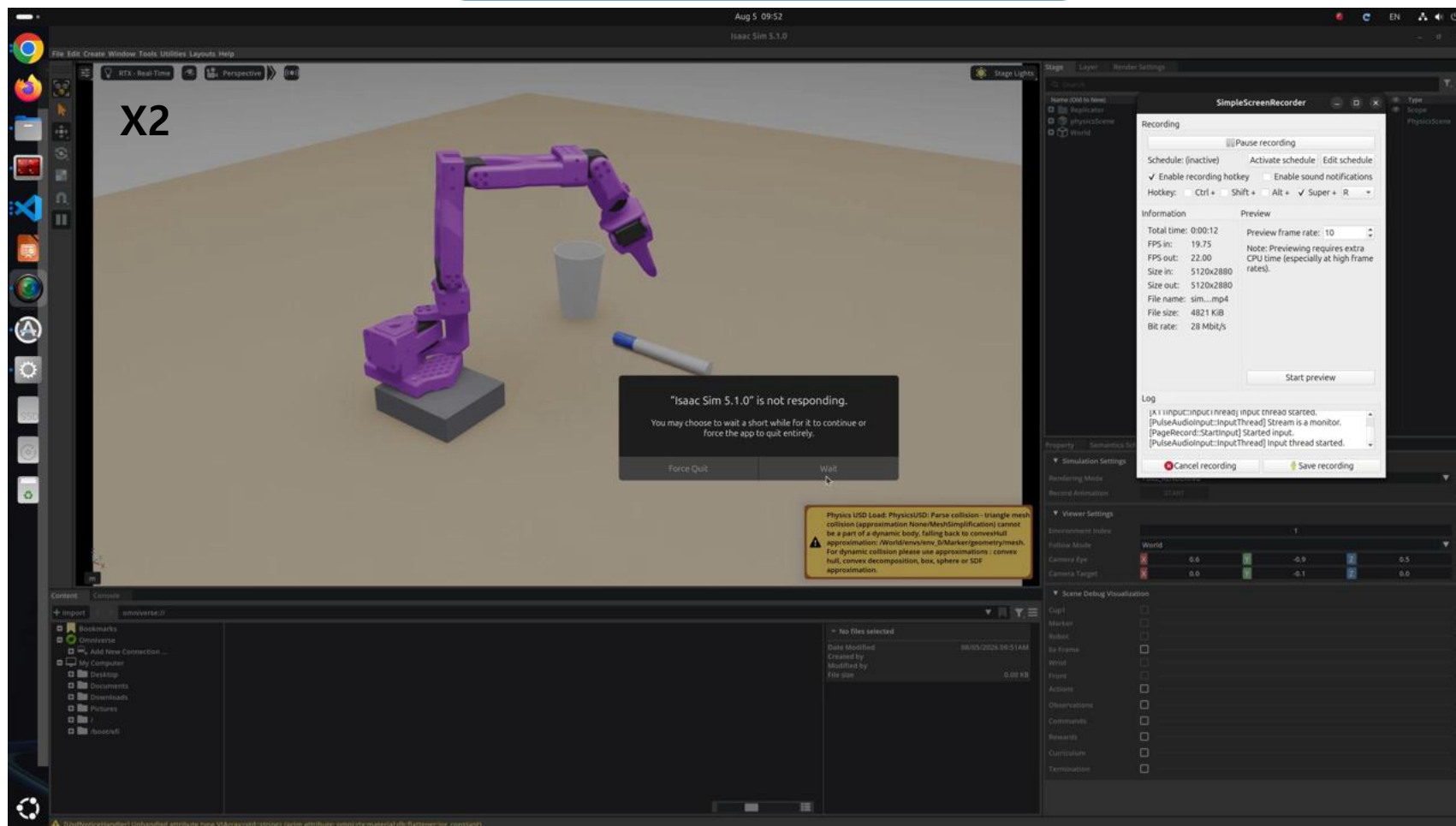
Experiment

100 dataset + action chunk=10

Success rate: 70%



Complete: 
 In-progress: 
 To-do: 



VLA pipeline Experiment

Complete: 
 In-progress: 
 To-do: 

1st training

Data collection
49 episode, 7min

Training
LoRA, 30K step,
Action chunk =10

Sim-evaluation
Success rate: 10%

2nd training - 1

Data collection
100 episode, 17min

Training
LoRA, 30K step,
Action chunk =10

Sim-evaluation
Success rate: **70%**

3rd training - 1

Data collection
150 episode, 24min

Training
LoRA, 30K step,
Action chunk =30

Real-evaluation
Success rate: **33%**

Sim-evaluation
Success rate: **70%**

2nd training - 2

Data collection
100 episode, 17min

Training
LoRA, 30K step,
Action chunk =30

Real-evaluation
Success rate: **25%**

Sim-evaluation
Success rate: **70%**

3rd training - 2

Data collection
180 episode, 32min

Training
LoRA, 30K step,
Action chunk =30

Real-evaluation
Success rate: **??**

Sim-evaluation
Success rate: **60%**

2nd training - 3

Data collection
100 episode, 17min

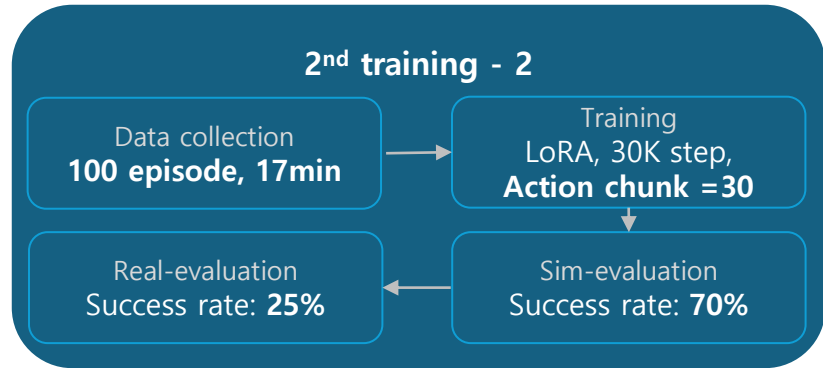
Training
LoRA, 30K step,
Action chunk =30
**Inference action
chunk = 10**

Sim-evaluation
Success rate: **50%**

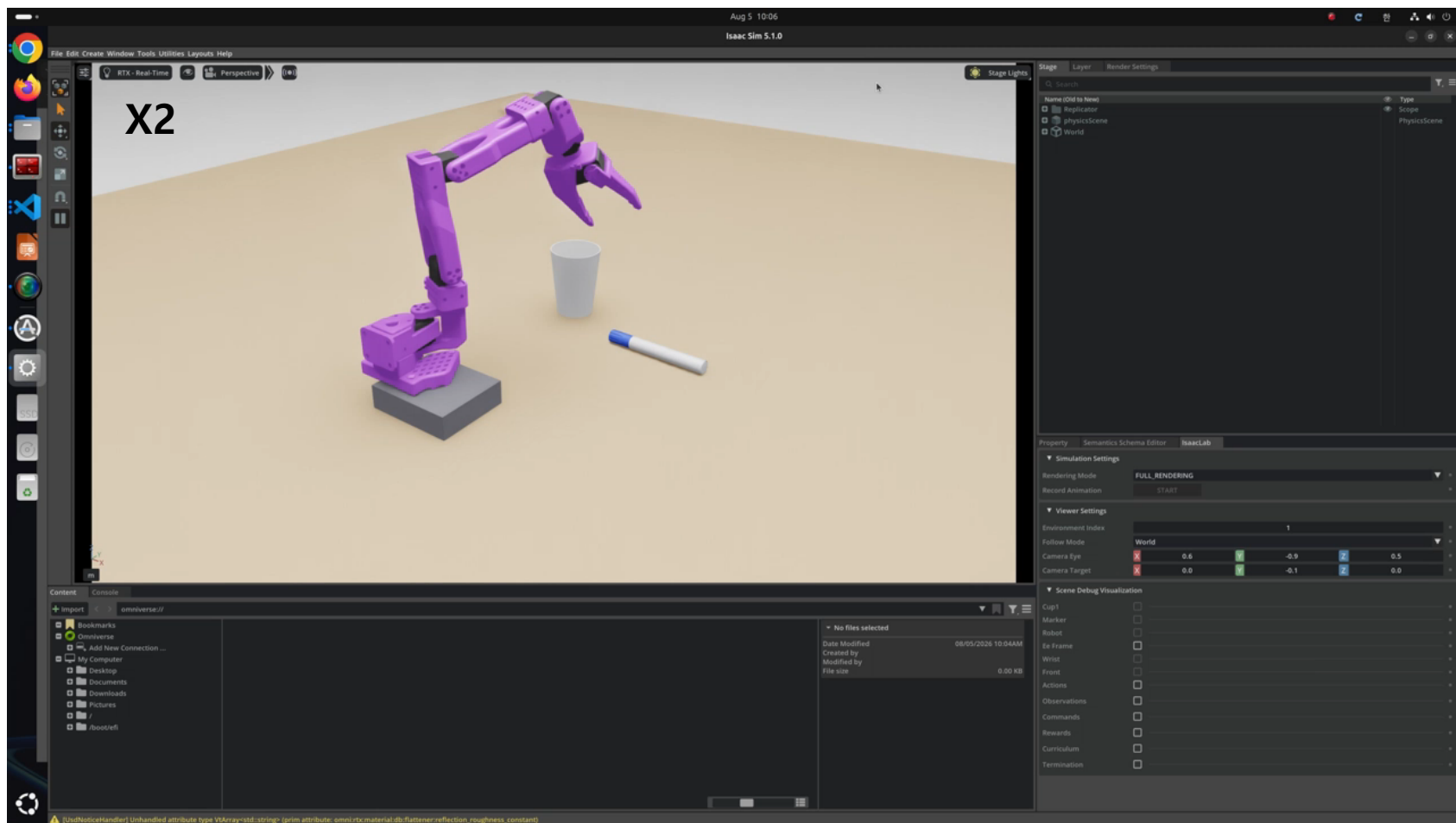
VLA pipeline Experiment

100 dataset + action chunk=30

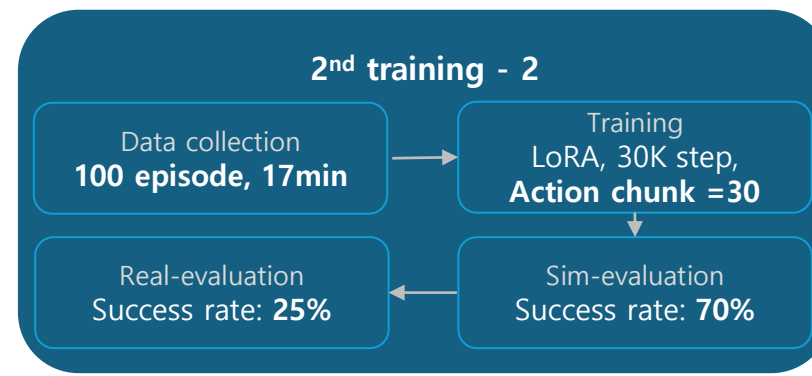
Success rate: 70%



Complete: 
 In-progress: 
 To-do: 

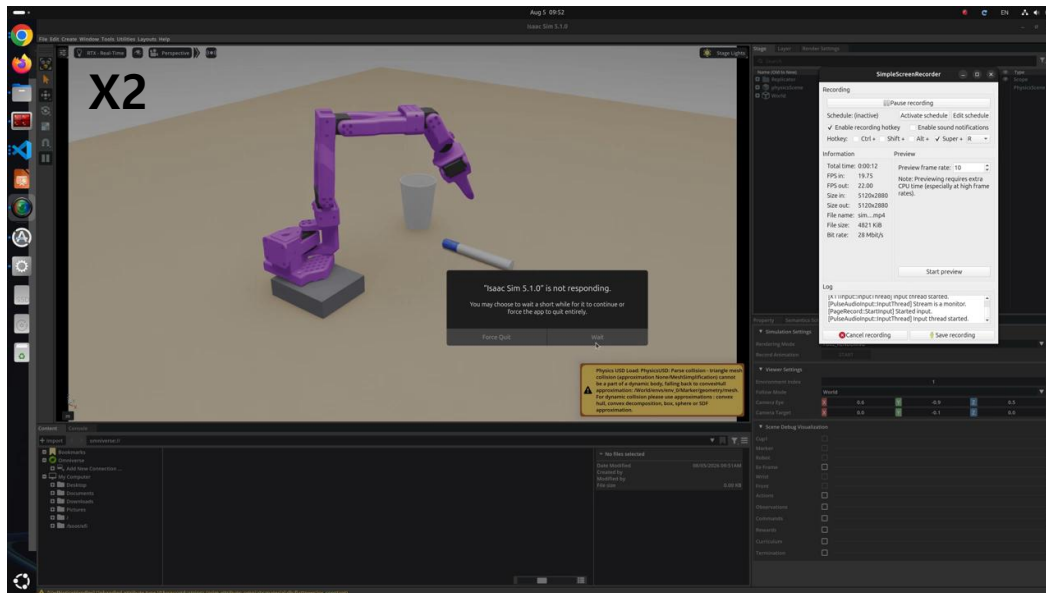


VLA pipeline Experiment

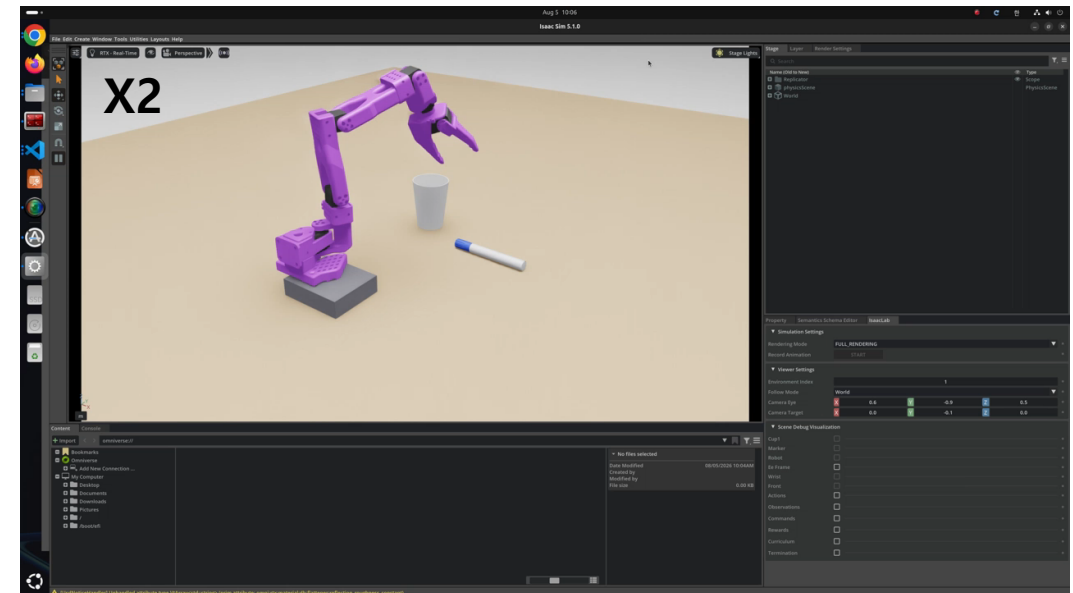


Complete: 
 In-progress: 
 To-do: 

action chunk=10



action chunk=30



VLA pipeline

Experiment_real

100 dataset + action chunk=30

Success rate: 25%

2nd training - 2

Data collection
100 episode, 17min

Training
LoRA, 30K step,
Action chunk =30

Real-evaluation
Success rate: 25%

Sim-evaluation
Success rate: 70%

Complete: 
In-progress: 
To-do: 



success



fail



fail

VLA pipeline Experiment

Complete: 
 In-progress: 
 To-do: 

1st training

Data collection
49 episode, 7min

Training
LoRA, 30K step,
Action chunk =10

Sim-evaluation
Success rate: 10%

2nd training - 1

Data collection
100 episode, 17min

Training
LoRA, 30K step,
Action chunk =10

Sim-evaluation
Success rate: **70%**

3rd training - 1

Data collection
150 episode, 24min

Training
LoRA, 30K step,
Action chunk =30

Real-evaluation
Success rate: **33%**

Sim-evaluation
Success rate: **70%**

2nd training - 2

Data collection
100 episode, 17min

Training
LoRA, 30K step,
Action chunk =30

Real-evaluation
Success rate: **25%**

Sim-evaluation
Success rate: **70%**

3rd training - 2

Data collection
180 episode, 32min

Training
LoRA, 30K step,
Action chunk =30

Real-evaluation
Success rate: **??**

Sim-evaluation
Success rate: **60%**

2nd training - 3

Data collection
100 episode, 17min

Training
LoRA, 30K step,
Action chunk =30
**Inference action
chunk = 10**

Sim-evaluation
Success rate: **80%**

VLA pipeline

Experiment

100 dataset + action chunk=30

Inference → action chunk=10

successrate : 80%

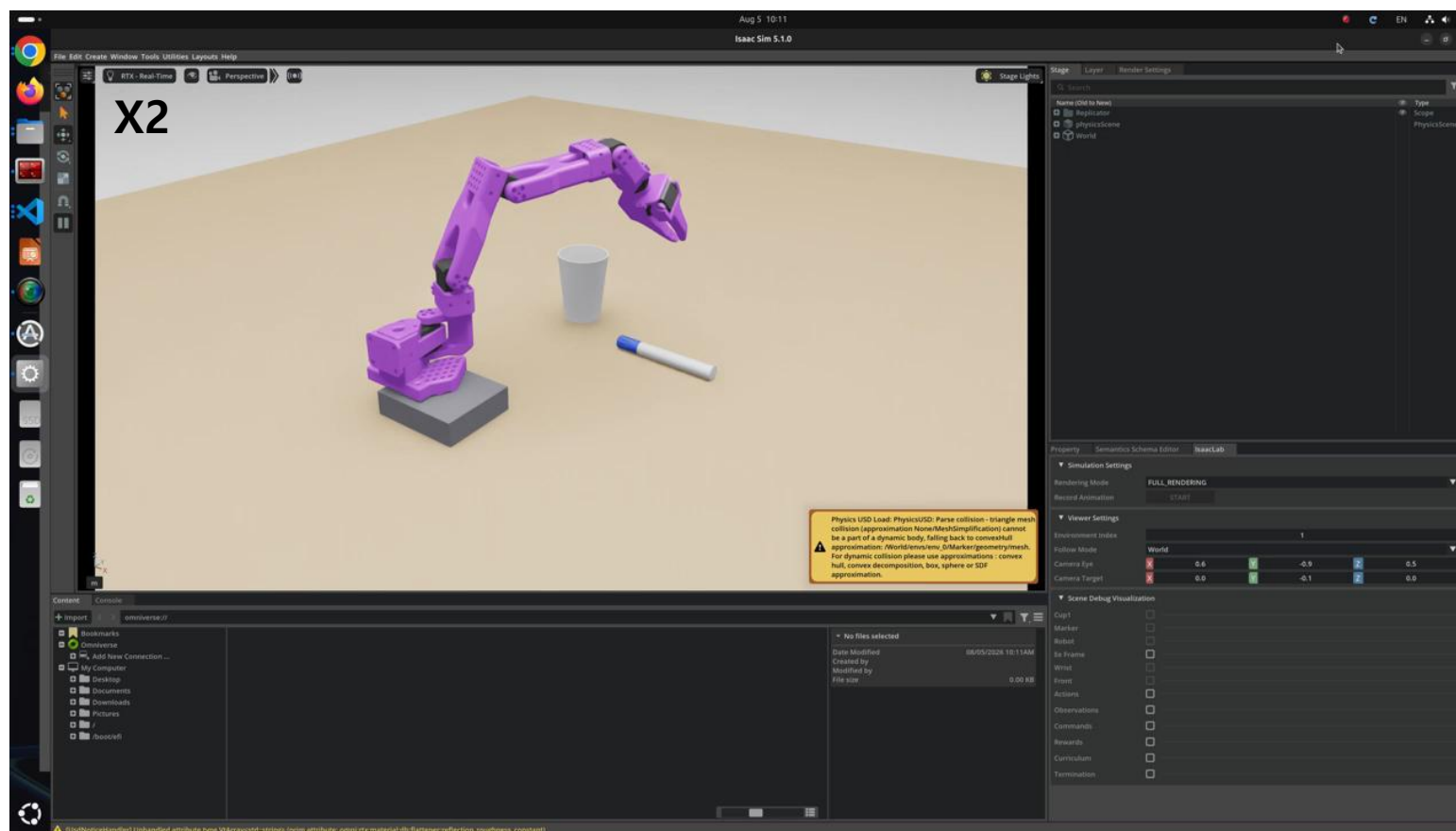
2nd training - 3

Data collection
100 episode, 17min

Sim-evaluation
Success rate:80%

Training
LoRA, 30K step,
Action chunk =30
Inference action
chunk = 10

Complete: 
In-progress: 
To-do: 



VLA pipeline Experiment

Complete: 
 In-progress: 
 To-do: 

1st training

Data collection
49 episode, 7min

Training
LoRA, 30K step,
Action chunk =10

Sim-evaluation
Success rate: 10%

2nd training - 1

Data collection
100 episode, 17min

Training
LoRA, 30K step,
Action chunk =10

Sim-evaluation
Success rate: **70%**

3rd training - 1

Data collection
150 episode, 24min

Training
LoRA, 30K step,
Action chunk =30

Real-evaluation
Success rate: **33.3%**

Sim-evaluation
Success rate: **63.3%**

2nd training - 2

Data collection
100 episode, 17min

Training
LoRA, 30K step,
Action chunk =30

Real-evaluation
Success rate: **25%**

Sim-evaluation
Success rate: **70%**

3rd training - 2

Data collection
180 episode, 32min

Training
LoRA, 30K step,
Action chunk =30

Real-evaluation
Success rate: **25%**

Sim-evaluation
Success rate: **70%**

2nd training - 3

Data collection
100 episode, 17min

Training
LoRA, 30K step,
Action chunk =30
**Inference action
chunk = 10**

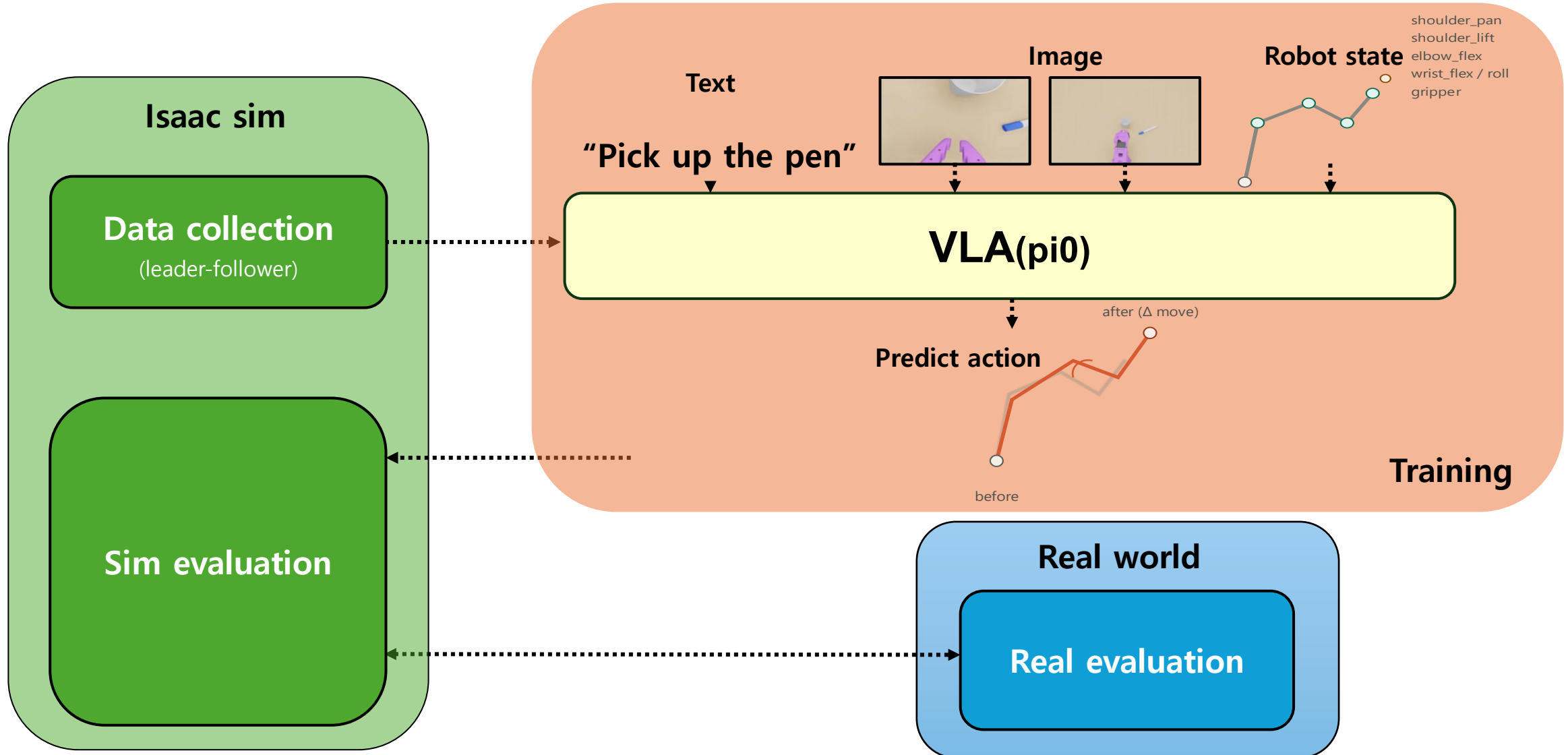
Sim-evaluation
Success rate: **33%**

#	Name	Dataset	Training H	Inference H	50ep	30ep
					Success rate (sim)	Success rate (real)
1	50ep_H10	Sim 49ep	10	10	10%	0%
2	100ep_H10	Sim 97ep	10	10	80%	30%
3	100ep_H30	Sim 97ep	30	30	66.7%	25%
4	100ep_H30_exec10	Sim 97ep	30	10	80%	25%
5	150ep_H30	Sim 146ep	30	30	63.3%	33%
6	180ep_H30_real	Sim 146ep + real 34ep	30	30	70%	25%

H = action chunk length

VLA Pipeline

Pi0 pipeline



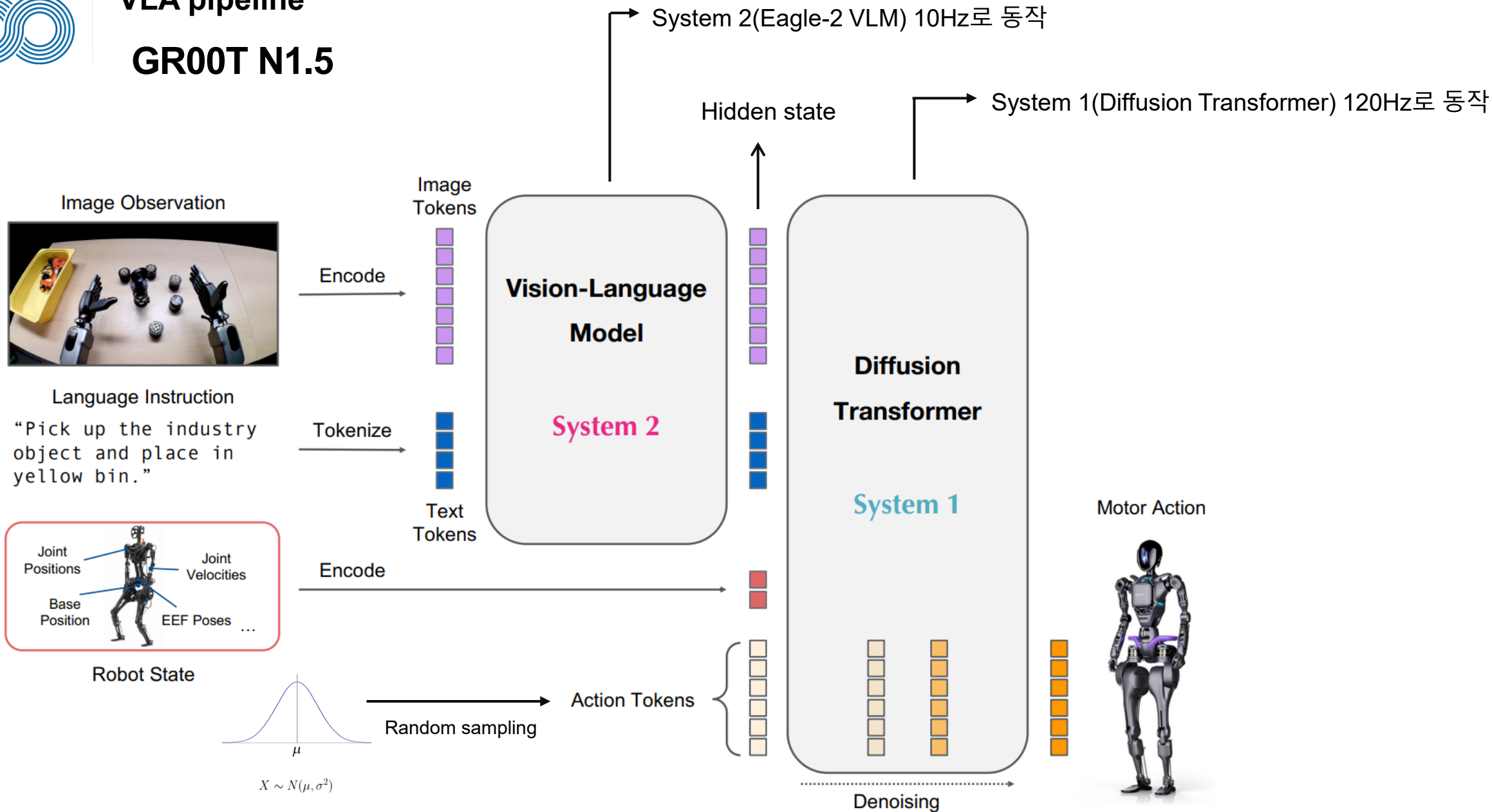
CHAPTER

02

GR00T N1.5

VLA pipeline

GR00T N1.5



Aspect	GR00T N1.5	$\pi 0$
구조	Dual-system (VLM + separate DiT)	단일 통합 모델 (PaliGemma + action expert)
VLM 학습 방식	Frozen	함께 fine-tune됨
VLM-action 연결	별도 DiT의 cross-attention	같은 Transformer 안의 self-attention
동작 주파수	System1 120Hz / System2 10Hz	단일 주파수로 통합 동작
주 타겟 로봇	휴머노이드 (양팔+손) 중심	다양한 embodiment (단일팔 포함)

GR00T N1.5: 물리적으로 분리해서 서로 다른 주파수로 돌림

$\pi 0$: 하나의 모델 안에서 통합

VLA pipeline

GR00T N1.5


Hugging Face

[Models](#)
[Datasets](#)
[Spaces](#)
[Buckets](#)
[NEW](#)
[Docs](#)
[Pricing](#)

Datasets:

mimiminsoo/marker_100
like 0

Tasks: [Robotics](#)
Modalities: [Tabular](#) [Time-series](#) [Video](#)
Formats: [parquet](#)
Size: [10K - 100K](#)
Tags: [LeRobot](#)

Libraries: [Datasets](#) [Dask](#) [Polars](#) + 2
 License: [apache-2.0](#)

[Dataset card](#)
[Data Studio](#)
[Files and versions](#)
[xet](#)
[Community](#) 1
 [Settings](#)

Dataset Viewer
[Auto-converted to Parquet](#)
[API](#)
[Embed](#)
[Duplicate](#)
[Data Studio](#)

Split (1)
train · 31.3k rows

Search is not available for this dataset

action list	observation.state list	timestamp float32	frame_index int64	episode_index int64	index int64
[21.78931427001...	[22.73485565185...	0	0	0	0
[21.78931427001...	[22.80239868164...	0.033333	1	0	0
[21.78931427001...	[22.41656494140...	0.066667	2	0	0

[→ Copy to bucket](#)
[NEW](#)
[Use this dataset](#)

[Edit dataset card](#)

Downloads last month **115**

License:
apache-2.0

Number of rows:
31,291

Total file size:
143 MB

VLA pipeline

GR00T N1.5 sim eval

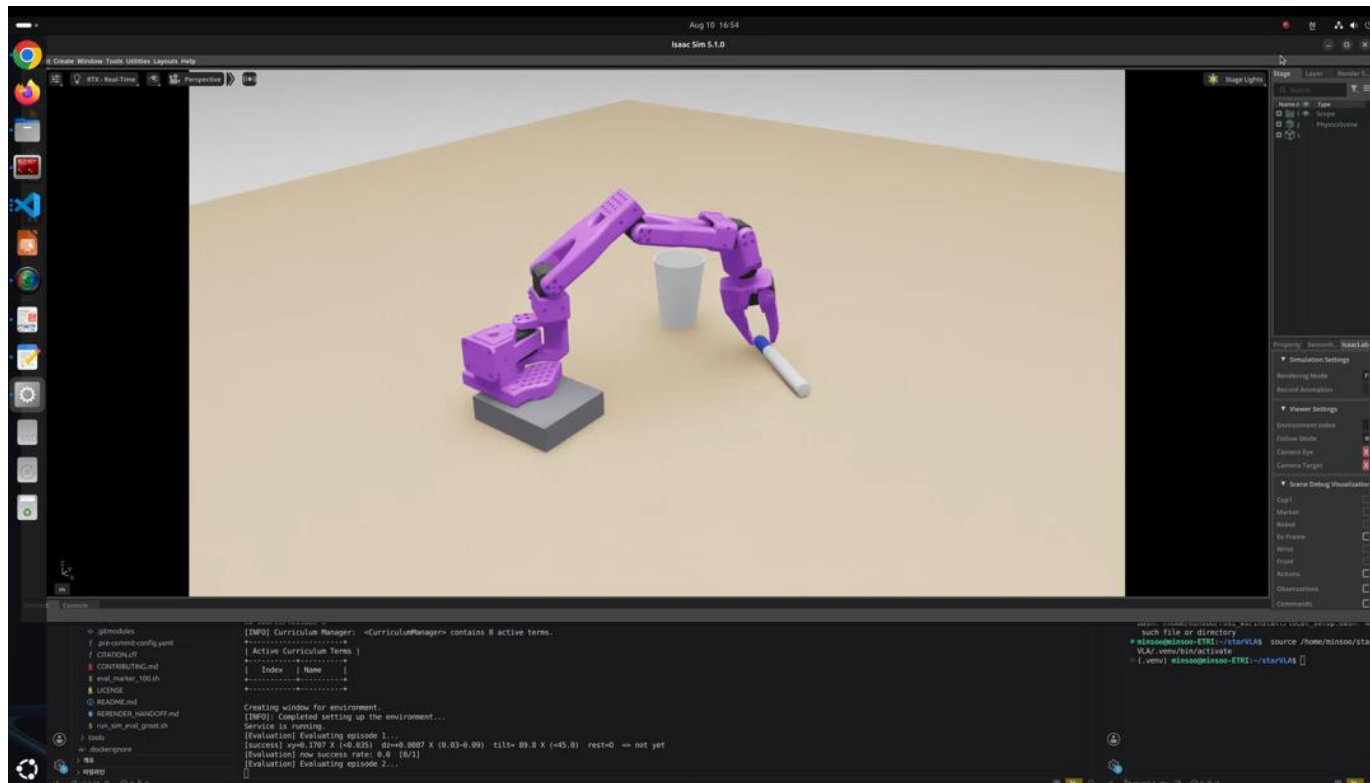
100 dataset

+ action chunk=10

Successrate: 40%

원인 가능성

1. Pretrained dataset
2. Training setting
3. Preprocessing
4. Embodiment
5. ...

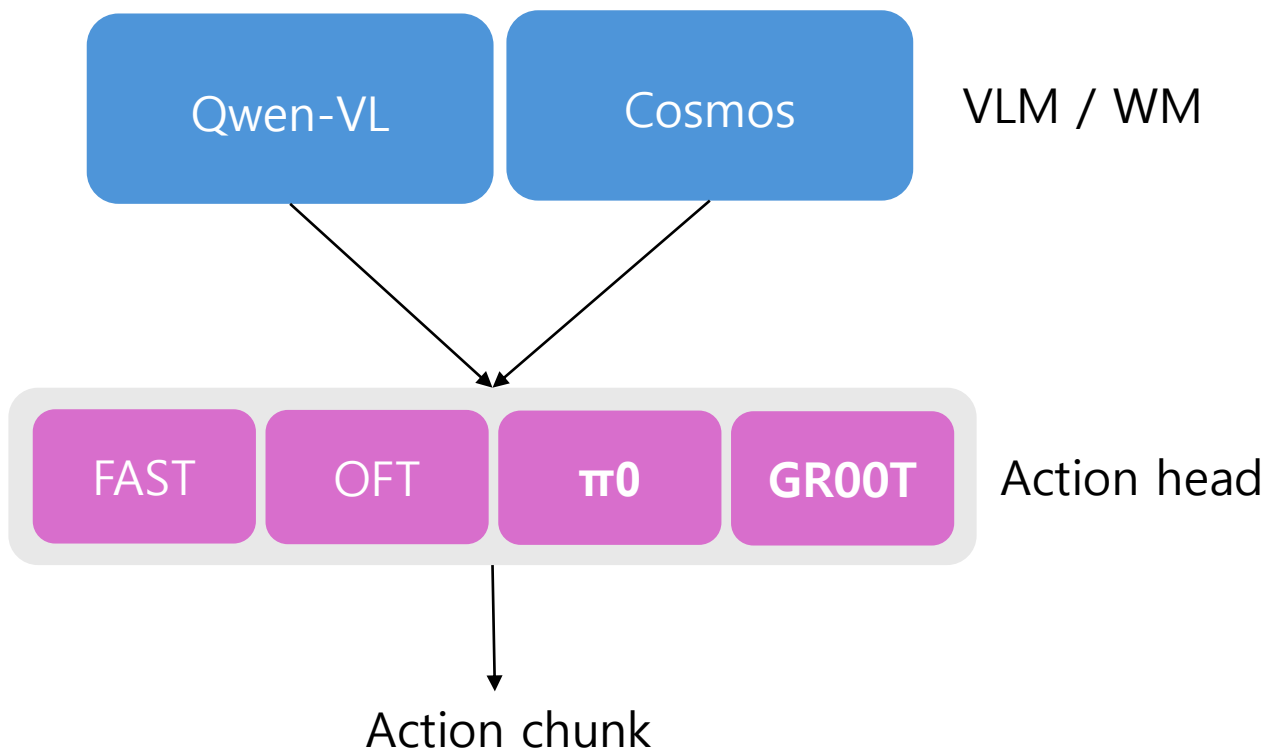
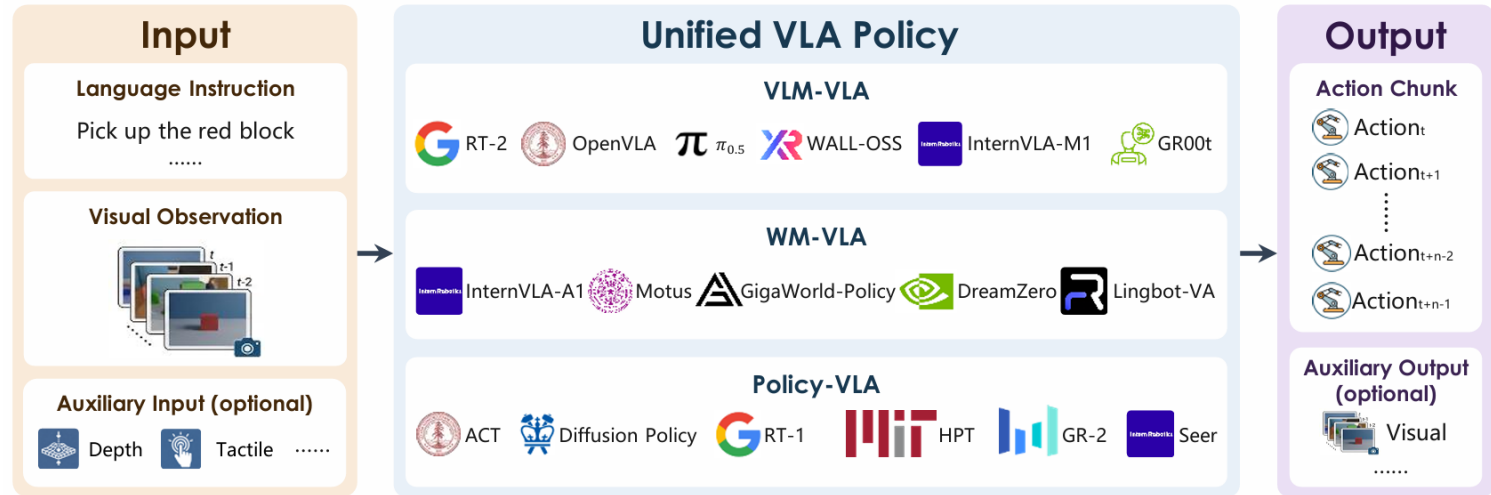


CHAPTER

03

starVLA

VLA pipeline starVLA core concept



- Same data
- Same training
- Same evaluation

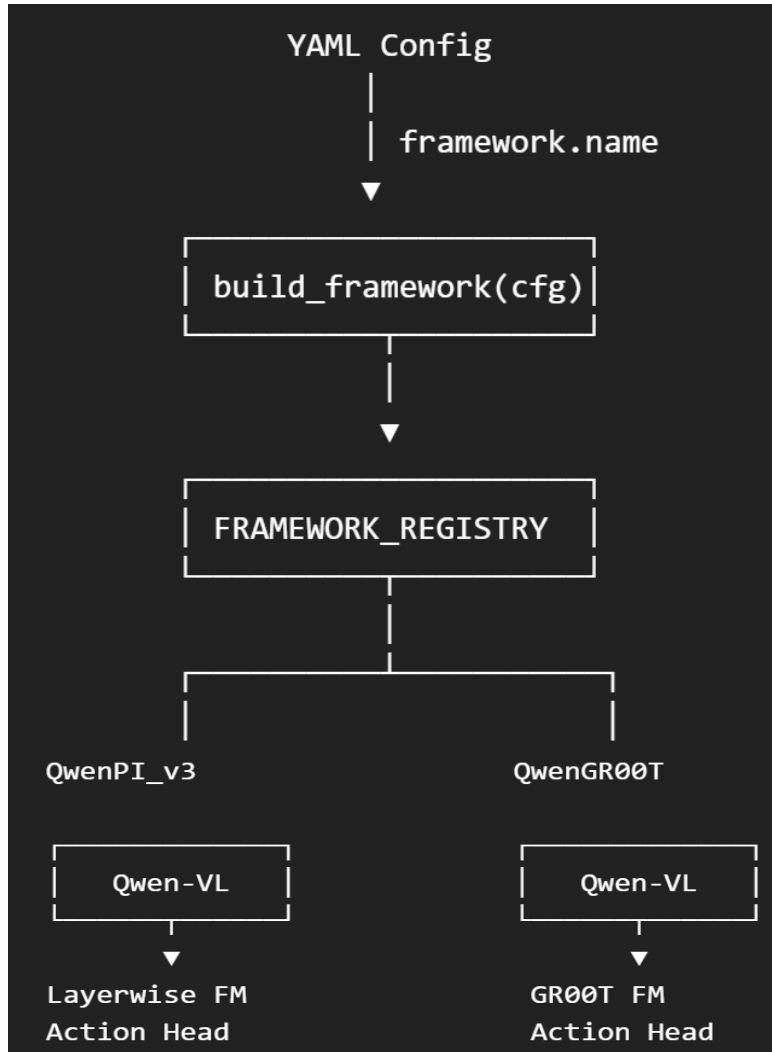
VLA pipeline

Class key

#	registry key	backbone	action head	paradigm
1	QwenOFT	Qwen (2.5/3-VL, base_vlm로 결정)	MLP_ActionHeader	OFT
2	QwenFast	Qwen	fast_ActionHeader	FAST
3	QwenPI	Qwen	LayerwiseFM_ActionHeader::LayerwiseFlowmatchingActionHead	π (StarVLA- π , 구버전)
4	QwenPI_v3	Qwen	LayerwiseFM_ActionHeader::LayerwiseFlowmatchingActionHead	π (StarVLA-π v3)
5	QwenGR00T	Qwen	GR00T_ActionHeader::FlowmatchingActionHead	GR00T
6	QwenDual	Qwen + DINO (dual vision encoder)	GR00T_ActionHeader::FlowmatchingActionHead	GR00T (dual-encoder 변형)
7	QwenDiscreteDiffusion	Qwen	LayerwiseDiscreteDiffusion_ActionHeader::LayerwiseDiscreteDiffusionActionHead	기타 (discrete diffusion)
8	QwenAdapter	Qwen	VLA_AdapterHeader::VLA_Adapter_L1RegressionActionHead	기타 (VLA-Adapter, L1 regression)
9	Gemma4GR00T	Gemma-4 (E2B)	GR00T_ActionHeader::FlowmatchingActionHead	GR00T
10	Gemma4PI	Gemma-4 (E2B)	LayerwiseFM_ActionHeader	π
11	MiniCPMGR00T	MiniCPM-V 4.6	GR00T_ActionHeader::FlowmatchingActionHead	GR00T
12	MiniCPMPI	MiniCPM-V 4.6	LayerwiseFM_ActionHeader	π
13	CosmosGR00T	Cosmos-Reason2 (NVIDIA, Qwen3-VL 아키텍처)	GR00T_ActionHeader::FlowmatchingActionHead	GR00T
14	LangForce	Qwen	GR00T_ActionHeader::FlowmatchingActionHead	GR00T 변형
15	ABot_M0	Qwen	AML_ActionHeader::FlowmatchingActionHead (GR00T와 별개 모듈)	기타 (자체 flow-matching)
16	EgoVLA	VILA (SigLIP-384 + Qwen2-1.5B)	EgoVLA_ActionHeader	기타 (EgoVLA 궤적 디코더)
17	InternVLA-M1	Qwen + DINO + QFormer projector	DiTActionHeader	기타 (InternVLA-M1 자체 DiT)
18	PI0 / Pi0(별칭)	PaliGemma (SigLIP + Gemma-2B), openpi 이식	OpenPI_ActionHead::OpenPI0ActionHead	original PI0
19	PI05 / Pi05(별칭)	PaliGemma, openpi 이식	OpenPI_ActionHead::OpenPI05ActionHead (AdaRMS 조건화)	original PI0 (π 0.5 변형)
20	CosmoPredict2OFT	Cosmos-Predict2 (world model)	MLP_ActionHeader	OFT
21	CosmoPredict2GR00T	Cosmos-Predict2	GR00T_ActionHeader::FlowmatchingActionHead	GR00T
22	CosmoPredict2PI	Cosmos-Predict2	LayerwiseFM_ActionHeader	π
23	WanOFT	Wan2.x (world model)	MLP_ActionHeader	OFT
24	WanGR00T	Wan2.x	GR00T_ActionHeader::FlowmatchingActionHead	GR00T
25	WanPI	Wan2.x	LayerwiseFM_ActionHeader	π
26	DiffusionPolicy	ResNet 비전 인코더 (VLM/LLM 없음, _dp_vendor/=real-stanford/diffusion_policy 벤더링)	DiffusionUnetImagePolicy (U-Net)	기타 (classic Diffusion Policy)
27	ACT	ResNet18 (실제 pip lerobot 패키지의 ACTPolicy optional import)	lerobot.policies.act.ACTPolicy	기타 (ACT, Zhao et al.)

VLA pipeline

How does StarVLA switch Action Heads?



① Config

framework:

name: **QwenPI_v3**

framework:

name: **QwenGR00T**

② Framework 생성

```
model = build_framework(cfg)
```

```
model_class = FRAMEWORK_REGISTRY[
    cfg.framework.name]
```

③ 내부 구조

```
# QwenPI_v3
```

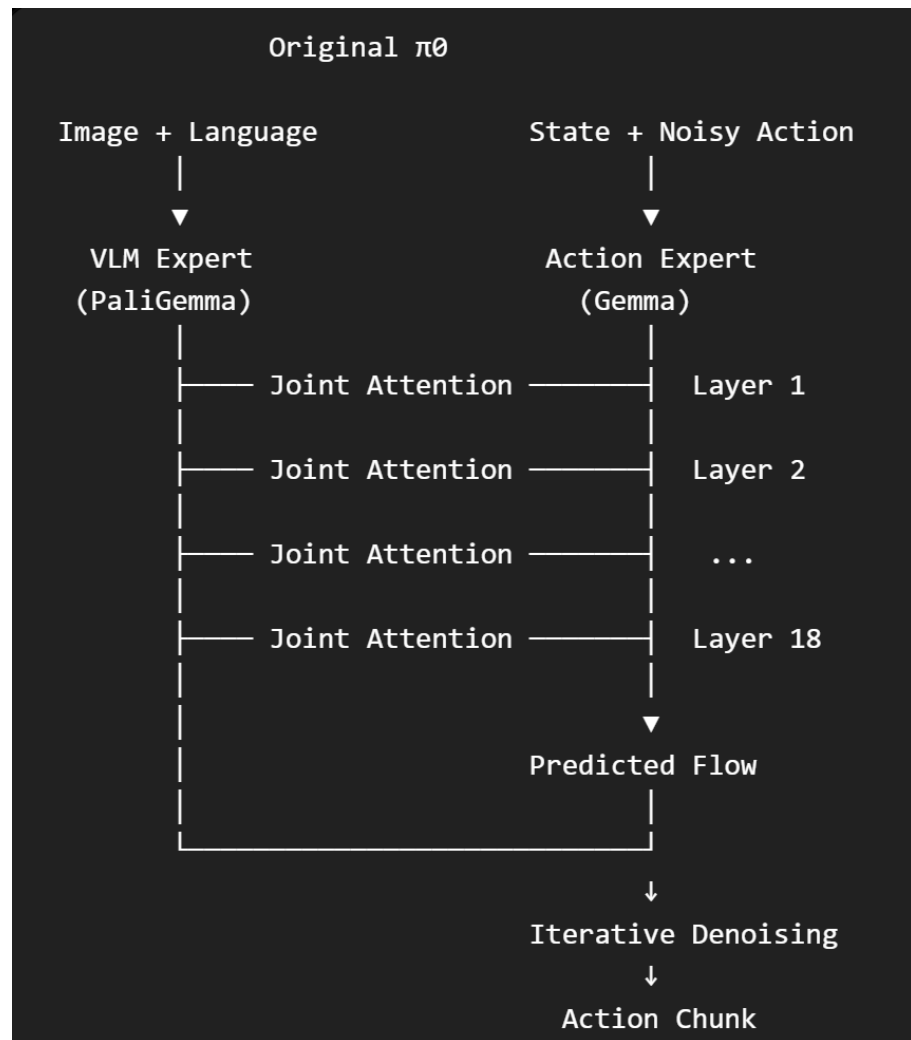
```
self.action_model =
    LayerwiseFlowmatchingActionHead(...)
```

```
# QwenGR00T
```

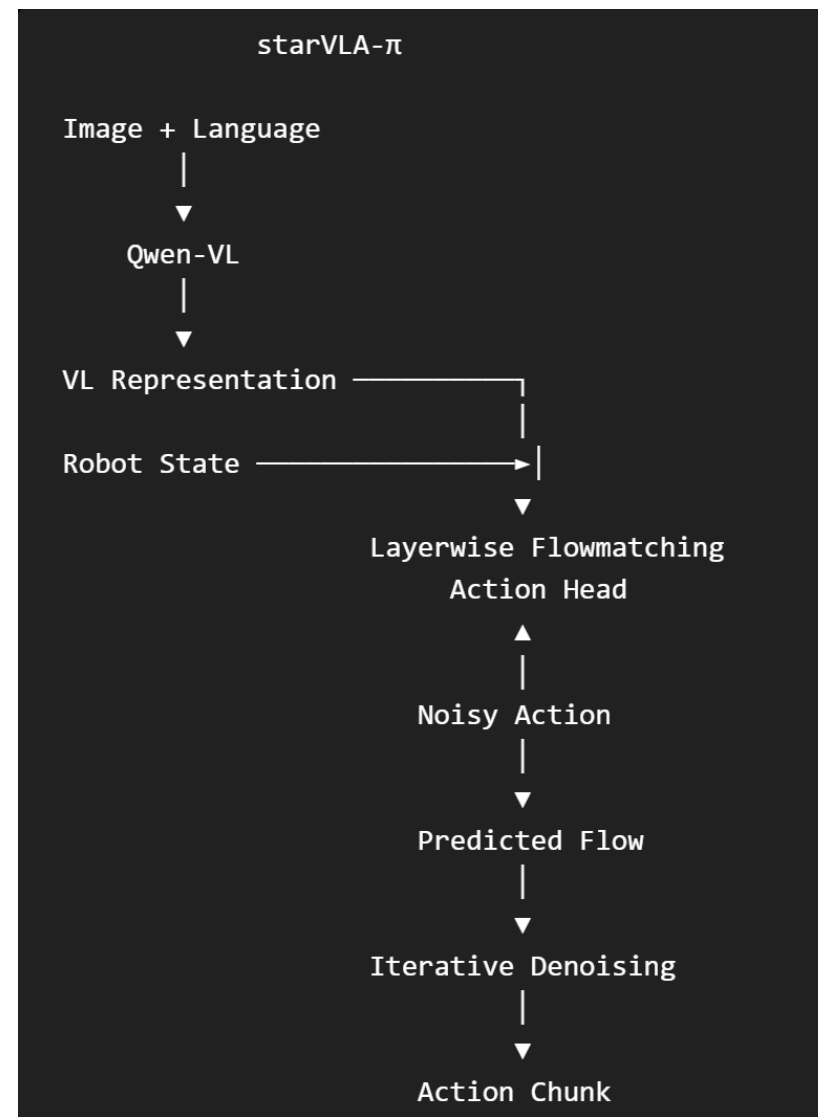
```
self.action_model =
    FlowmatchingActionHead(...)
```

VLA pipeline

Original π_0 vs. starVLA- π



Modulization



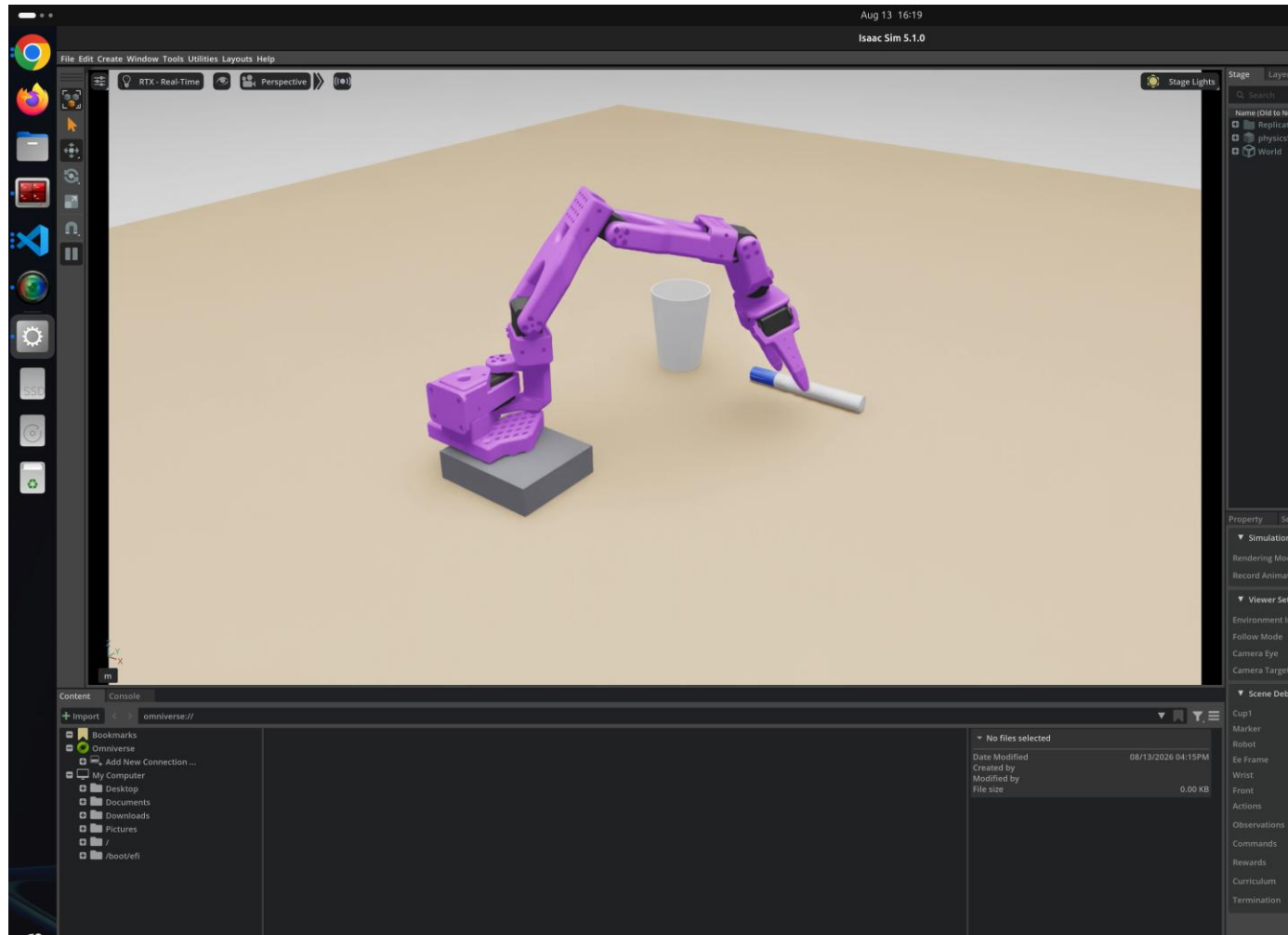
VLA pipeline

StarVLA- π vs StarVLA-GR00T

	StarVLA- π	StarVLA-GR00T
Backbone	Qwen	Qwen
Action generation	Flow Matching	Flow Matching
Action model	Layerwise Cross-DiT	GR00T-style DiT
VL conditioning	Multi-layer VL features	System-2 VL context
Interaction	layer-wise cross-attention	self-attention + VL cross-attention
설계 철학	VLM-action 깊은 결합	Reasoning-Action 분리

VLA pipeline

StarVLA pipeline validation



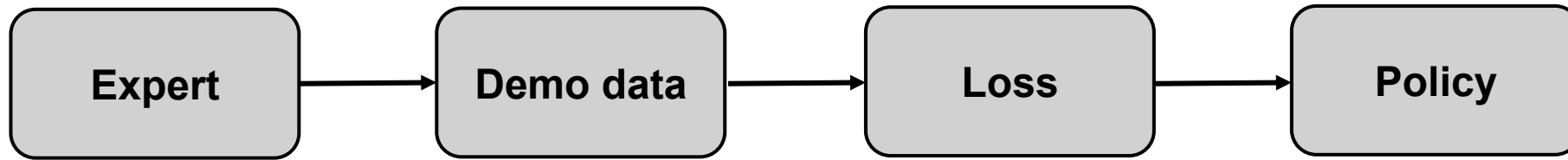
- 97demo
- 30K step training
- **Success rate: 0%**
- RTX 3090
 - Batch size=1

CHAPTER

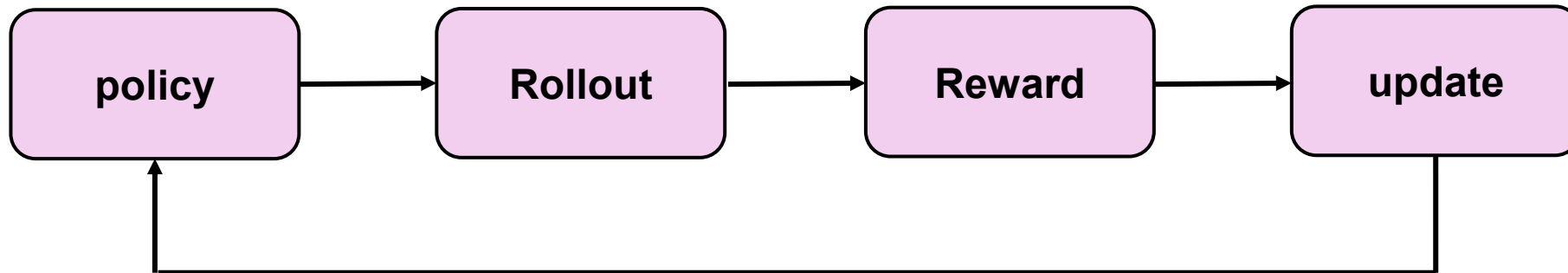
04

RL fine-tuning

SFT (supervised fine-tuning)

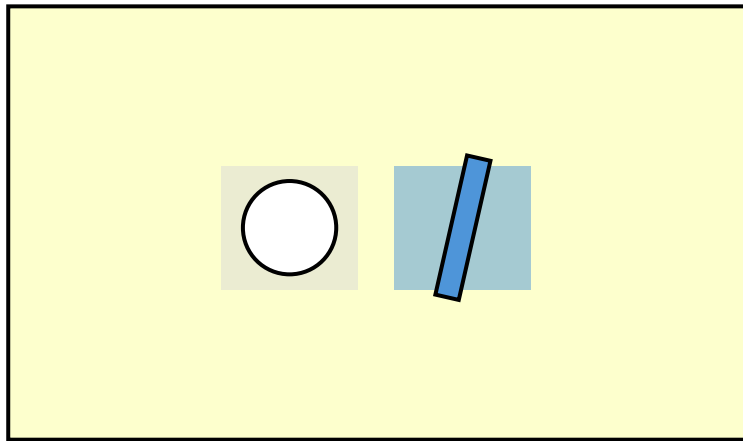


RL

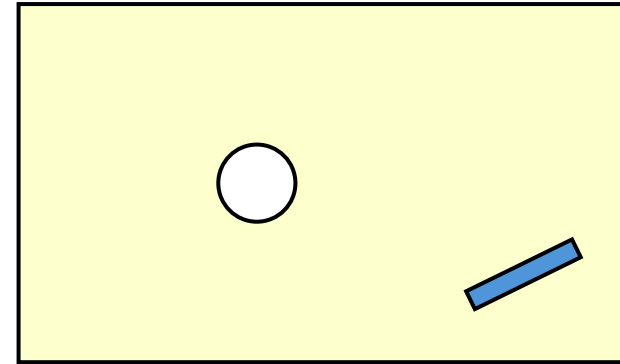


Why RL?

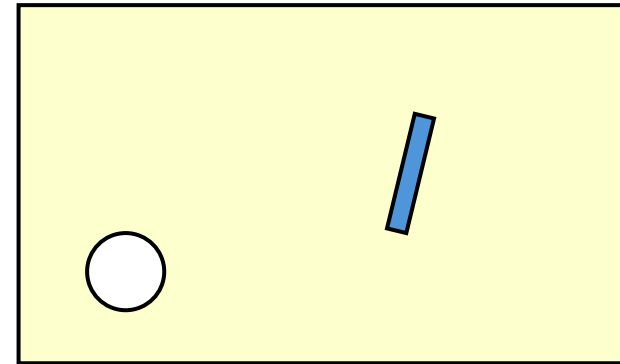
Expert demonstrations cannot cover all OOD states



Expert demonstrations



Unseen state 1



Unseen state 2

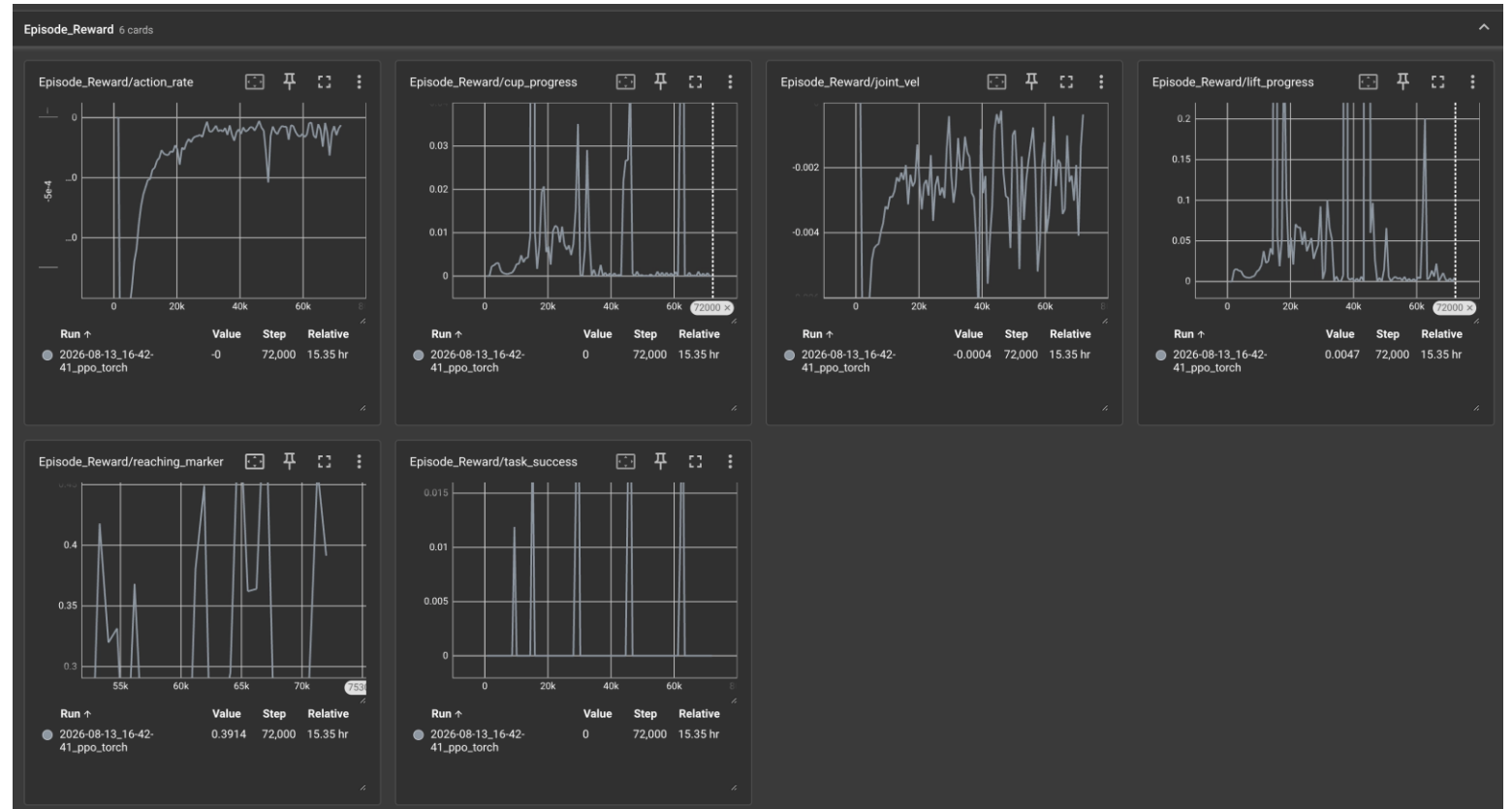
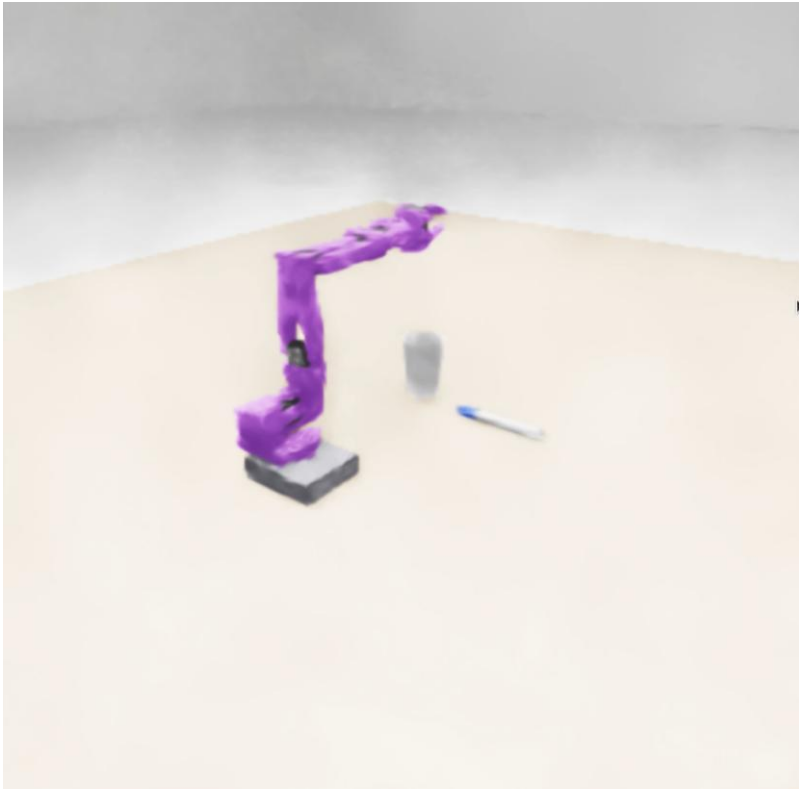
RL reduces expert data cost and improves robustness.

RL from scratch

Reward	역할	Weight
reaching_marker	EE가 marker에 접근	1.0
lifting_marker	marker 들어올리기	8.0
marker_cup_tracking	cup 방향으로 coarse 이동	8.0
marker_cup_tracking_fine	cup 근처 정밀 접근	4.0
task_success	최종 성공	60.0
action_rate	급격한 action 변화 억제	-1e-4
joint_vel	과도한 joint velocity 억제	-1e-4

RL fine-tuning Evaluation

PPO → 72K training

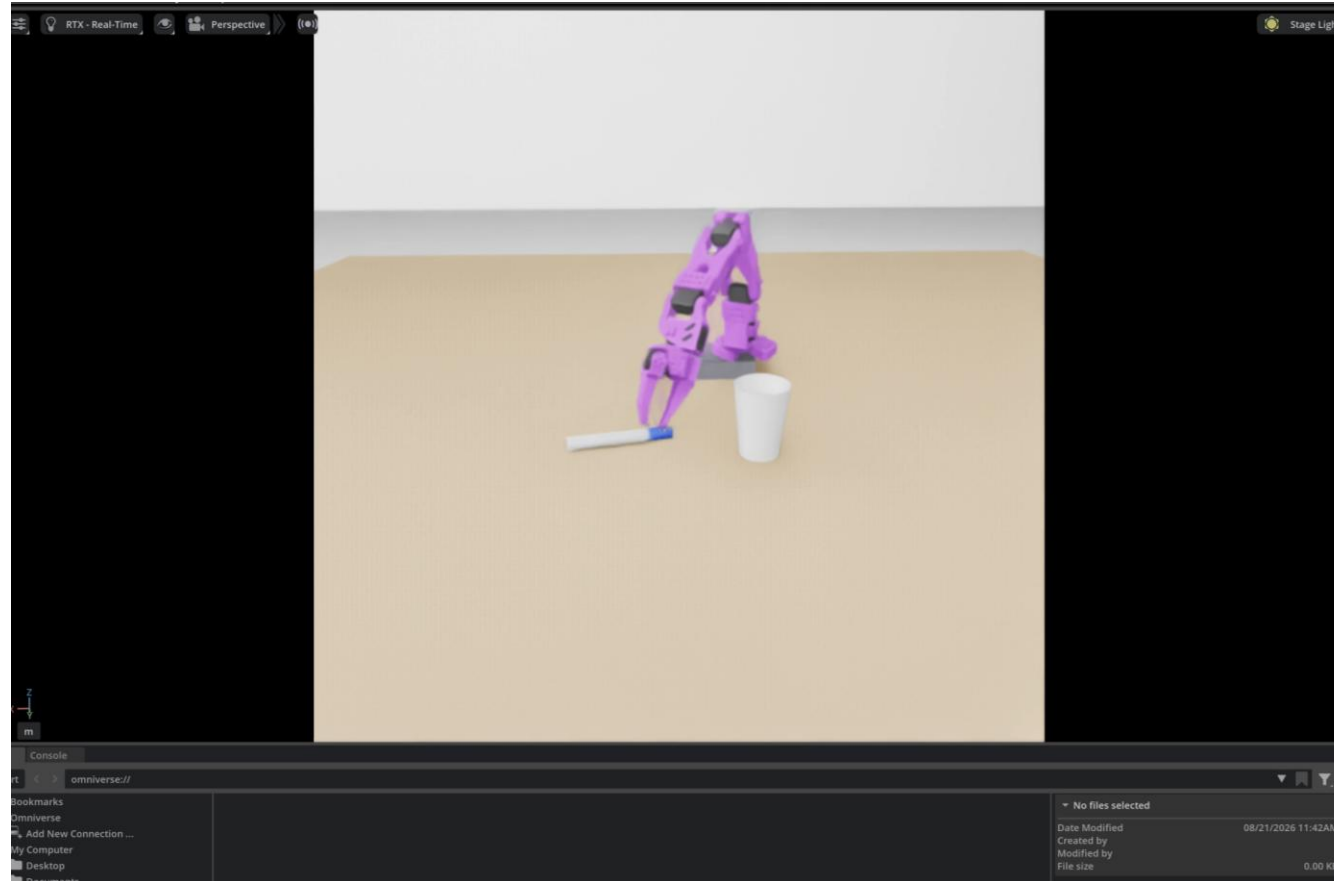


RL from scratch

Reward	역할	Weight
reach_progress	EE가 marker에 접근한 Δ만큼 보상	+2.0
secure_grasp_event	marker를 안정적으로 grasp한 순간 보상	+3.0
lift_progress	grasp 후 marker를 들어올린 Δ 만큼 보상	+4.0
safe_lift_event	marker를 cup rim보다 안전하게 들어올린 순간 보상	+2.0
above_cup_progress	marker가 cup 상단으로 XY 접근한 Δ 만큼 보상	+3.0
task_success	marker가 cup 안에 최종 안착	+480.0
time_penalty	빠른 task completion 유도	-0.03 / sec
action_rate	급격한 action 변화 억제	-1e-4
joint_vel	과도한 joint velocity 억제	-1e-4

RL fine-tuning Evaluation

PPO → 121K training



RL fine-tuning

GRPO → 900step training

Group size = 4

Num_env = 4



RTX 3090 → 9H

	π RL (Reference)	Me
GPU	H100 X 8 (총 640GB)	RTX 3090 X 1 (총 24GB)
Global batch size	2048~5120	4



What I Learned

❖ Model

- VLA architecture / flow matching / action chunk
- $\pi 0$, GR00T 등 foundation model 이해

❖ System

- Isaac Sim → dataset → training → inference → real robot
- simulator/model/robot 간 interface 중요성

❖ Research

- 높은 sim success가 real robustness를 보장하지 않음
- dataset quantity뿐 아니라 **coverage**와 **failure recovery**가 중요
- 실험 결과로부터 다음 연구 질문을 만드는 과정

Q & A

Thank you